

**Bounds on mixing of passive scalars advected by incompressible energy and
enstrophy constrained flows and Anomalous dissipation**

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Chapter 1

Introduction

Mixing is the reduction of in-homogeneity in order to achieve a homogeneous state, generally this means achieving a uniform distribution of the mixtures. The in-homogeneity can be temperature or concentration etc. Mixing is important in many aspects of science and engineering, and understanding of the underlying kinematics of mixing is important because it helps in building and optimization of mixing devices.

The following quantities are used to describe the degree of mixing:

1. intensity of segregation = variation in concentration
 - Homogenization is mixing by the reduction of intensity of segregation.
2. scale of segregation = distribution of length scales.
 - Filamentation is mixing by the reduction of scale of segregation.

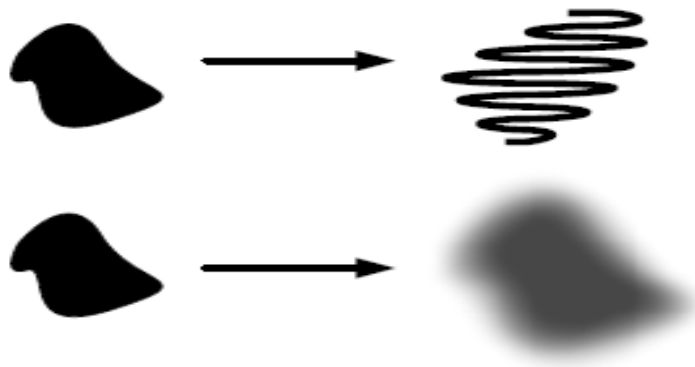


Figure 1.1: The top transformation illustrates filamentation as the reduction of the scale of segregation and the second illustrates homogenisation as the reduction of the intensity of segregation.

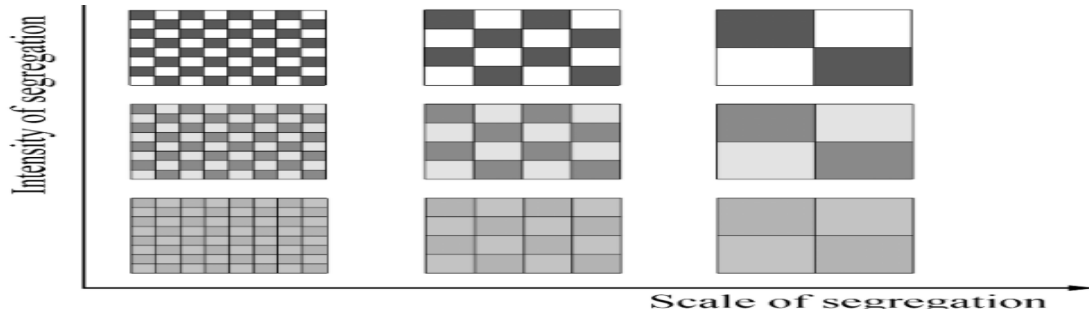


Figure 1.2: Taken from “What does well mixed mean?” by Prof. Suzanne Kresta, University of Alberta, San Francisco, 2013

Variance only measures the concentration variance or the intensity of segregation - not the arrangement of the areas. This requires a second measure: the scale of segregation.

1.1 Mathematical Interpretation

A mixed state is a state where the concentration is equal to the mean concentration. Intensity of segregation can be interpreted as the normalised variance of the concentration. It can be related to the L_2 norm of the concentration field. If $\theta(x; t)$ is the concentration of a passive scalar such as temperature, then the variance is given by

$$\text{var } \theta = ||\theta||^2 - \langle \theta \rangle^2 \quad (1.1.1)$$

where $||\theta||^2 = \frac{1}{|\Omega|} \int_{\Omega} \theta^2 \, d\Omega$, $\langle \theta \rangle = \frac{1}{|\Omega|} \int_{\Omega} \theta \, d\Omega$ are the L_2 - norm and mean value of, respectively, with the spatial domain Ω and $|\Omega|$ its volume or area in 2d.

1.2 Variance as a measure of mixing

This section illustrates how variance is used to measure the rate of mixing in the sense of advection and diffusion.

Now consider the advection-diffusion equation

$$\frac{\partial \theta}{\partial t} + u \cdot \nabla \theta = k \Delta \theta \quad (1.2.1)$$

$$\nabla \cdot u = 0 \quad \text{on } \Omega \quad (1.2.2)$$

with boundary conditions:

$$\vec{n} \cdot \nabla \theta = 0 \quad \text{on } \partial \Omega \quad (1.2.3)$$

$$\vec{n} \cdot \theta = 0 \quad \text{on } \partial \Omega \quad (1.2.4)$$

Multiply (1.2.1) by θ

$$\theta \frac{\partial \theta}{\partial t} + \theta u \cdot \nabla \theta = k \theta \Delta \theta$$

$$\begin{aligned}\frac{1}{2} \frac{\partial \theta^2}{\partial t} + \theta u \cdot \nabla \theta &= k \theta \nabla \cdot \nabla \theta \\ \frac{1}{2} \frac{\partial \theta^2}{\partial t} + \theta u \cdot \nabla \theta &= k \nabla \cdot (\theta \nabla \theta) - k \nabla \theta \cdot \nabla \theta \\ \frac{1}{2} \frac{\partial \theta^2}{\partial t} + \theta u \cdot \nabla \theta &= k \nabla \cdot (\theta \nabla \theta) - k |\nabla \theta|^2\end{aligned}$$

Integrating

$$\begin{aligned}\frac{d}{dt} \frac{1}{2} \int_{\Omega} \theta^2 \, d\Omega + \int_{\Omega} \theta u \cdot \nabla \theta \, d\Omega &= k \int_{\Omega} \nabla \cdot (\theta \nabla \theta) \, d\Omega - k \int_{\Omega} |\nabla \theta|^2 \, d\Omega \\ \frac{d}{dt} \frac{1}{2} \int_{\Omega} \theta^2 \, d\Omega &= -k \int_{\Omega} |\nabla \theta|^2 \, d\Omega\end{aligned}$$

Therefore

$$\frac{d}{dt} \|\theta\|^2 = -2k \|\nabla \theta\|^2$$

Integrating (1.2.1) over Ω

$$\begin{aligned}\int_{\Omega} \frac{\partial \theta}{\partial t} \, d\Omega + \int_{\Omega} u \cdot \nabla \theta \, d\Omega &= \int_{\Omega} k \Delta \theta \, d\Omega \\ \frac{d}{dt} \int_{\Omega} \theta \, d\Omega + \int_{\Omega} u \cdot \nabla \theta \, d\Omega &= \int_{\Omega} k \Delta \theta \, d\Omega\end{aligned}$$

Applying the boundary conditions, implies

$$\frac{d}{dt} \langle \theta \rangle = 0$$

, Using (1.1.1), then

$$\frac{d}{dt} \text{var} \theta = -2k \|\nabla \theta\|^2. \quad (1.2.5)$$

We can see that the right-hand side is negative-definite unless θ is constant. The concentration is driven to the uniform state $\text{var} \theta = 0$ at the rate $2k \|\nabla \theta\|_2^2$. *Therefore, studying the decay of the variance is an easy way of studying the effectiveness of a mixing process.*

1.3 Variance's Limitation

The physical interpretation of the (1.2.5) is that the stirring sharpens gradients of θ till they are big enough so that diffusion easily smooths them out. Then an agreement between advection and diffusion is then reached, and the variance decays at its optimal rate.

If we consider the case where diffusion is absent, that is, the case where $k = 0$, the

variance is conserved and does not change with time. Since diffusion has small effect compared to advection during the mixing process, it is indicative to study the advection equation.

$$\frac{\partial \theta}{\partial t} + u \cdot \nabla \theta = 0 \quad (1.3.1)$$

In this case, we can not use the variance, we will look for another measure of mixing!

1.4 Negative Sobolev Norm

Mathew, Mezic and Petzold [MMP05] introduced the mix-norm, which for mean-zero functions is equivalent to $\|\theta(\cdot, t)\|_{H^{-\frac{1}{2}}}$. Charles Doering and Lin. Thiffeault [Thi12] generalized the mix-norm to $\|\theta(\cdot, t)\|_{H^{-a}}$, $a > 0$.

Advantages

- The H^{-a} norm accounts for both scale and intensity of segregation.
- decays even in the absence of diffusion.

Definition 1.4.1. The negative Sobolev norm of θ is given as

$$\|\theta(\cdot, t)\|_{H^{-a}}^2 = \|\nabla|^{-a}\theta(\cdot, t)\|_{L^2}^2 = \sum_{\mathbf{k} \neq 0} k^{-2a} |\hat{\theta}_{\mathbf{k}}(t)|^2 \quad (a > 0)$$

where $k = |\mathbf{k}|$ and $\hat{\theta}_{\mathbf{k}}(t) = \frac{1}{L^{\frac{d}{2}}} \int_{[0, L]^d} e^{-i\mathbf{k}x} \theta(x, t) dx$

Definition 1.4.2. The stirring field $u(x, t)$ is called mixing if for every periodic square-integrable function $g(x)$ on $[0, L]^d$,

$$\lim_{t \rightarrow \infty} \int_{[0, L]^d} g(x) \theta(x, t) dx = \langle \theta \rangle$$

Theorem 1 ([Thi12, Theorem 1]). *Suppose the spatially mean-zero function $\theta(x, t)$ is bounded uniformly in $L^2([0, L]^d)$ for all $t > 0$. Then*

$$\begin{aligned} \lim_{t \rightarrow \infty} \int_{[0, L]^d} g(x) \theta(x, t) dx &= 0 \quad \forall g \in L^2 \\ \iff \lim_{t \rightarrow \infty} \|\theta(\cdot, t)\|_{H^{-a}} &= 0 \end{aligned}$$

for any $a > 0$

Proof. Assume that $\theta(\cdot, t)$ is uniformly bounded in $L^2([0, L]^d)$, that is $|\theta(\cdot, t)|_{L^2} \leq C$, and $\lim_{t \rightarrow \infty} \|\theta(\cdot, t)\|_{H^{-a}} = 0$ for some $a > 0$. Then for any $g \in L^2([0, L]^d)$,

Using parseval inequality, we have

$$\begin{aligned} \left| \int_{[0,L]^d} \theta g d\vec{x} \right| &= \left| \sum_k \hat{\theta}_{\vec{k}} \hat{g}_{\vec{k}}^* \right| \\ &= \left| \sum_{k \leq K} k^{-a} \hat{\theta}_{\vec{k}} k^a \hat{g}_{\vec{k}}^* + \sum_{k > K} \hat{\theta}_{\vec{k}} \hat{g}_{\vec{k}}^* \right| \end{aligned}$$

Using triangle inequality and cauchy-schwartz inequality, we have

$$\leq |\theta|_{H^{-a}} \left(\sum_{k \leq K} k^{2a} |\hat{g}_{\vec{k}}|^2 \right)^{1/2} + |\theta|_{L^2} \left(\sum_{k > K} |\hat{g}_{\vec{k}}|^2 \right)^{1/2}.$$

where $\hat{\theta}_{\vec{k}}$ and $\hat{g}_{\vec{k}}^*$ are the fourier coefficients of θ and g respectively. Now let $\varepsilon > 0$, and choose $K(\varepsilon)$ such that $(\sum_{k > K(\varepsilon)} |\hat{g}_{\vec{k}}|^2)^{1/2} \leq \varepsilon/2C$, then choose $N(\varepsilon)$ such that $|\theta(\cdot, N(\varepsilon))|_{H^{-a}} \leq \frac{1}{2}\varepsilon (\sum_{k \leq K(\varepsilon)} k^{2a} |\hat{g}_{\vec{k}}|^2)^{-1/2}$, for $t > N(\varepsilon)$. Then

$$\begin{aligned} \left| \int_{[0,L]^d} \theta g d\vec{x} \right| &\leq \frac{1}{2}\varepsilon \left(\sum_{k \leq K(\varepsilon)} k^{2a} |\hat{g}_{\vec{k}}|^2 \right)^{-1/2} \left(\sum_{k \leq K} k^{2a} |\hat{g}_{\vec{k}}|^2 \right)^{1/2} + |\theta|_{L^2} \varepsilon/2C. \\ &\leq \frac{1}{2} (1 + C^{-1} |\theta|_{L^2}) \varepsilon \leq \varepsilon, \quad t > N(\varepsilon). \end{aligned}$$

$$\left| \int_{[0,L]^d} \theta g d\vec{x} \right| \leq \frac{1}{2} (1 + C^{-1} |\theta|_{L^2}) \varepsilon \leq \varepsilon, \quad t > N(\varepsilon), \quad (1.4.1)$$

$\implies \theta$ converges weakly to zero as $t \rightarrow \infty$.

Conversely, assume $|\theta(\cdot, t)|_{L^2} \leq C$ for all t and $\lim_{t \rightarrow \infty} \int_{[0,L]^d} \theta g d\vec{x} \rightarrow 0$ for all $g \in L^2([0,L]^d)$. Choose $g = \exp(-i\vec{k} \cdot \vec{x})$ we can see that the Fourier coefficients $\hat{\theta}_{\vec{k}}(t) \rightarrow 0$ as $t \rightarrow \infty$.

Since $|\theta(\cdot, t)|_{L^2}^2 = \sum_{\vec{k}} |\hat{\theta}_{\vec{k}}(t)|^2 \leq C^2$, each $|\hat{\theta}_{\vec{k}}(t)| \leq C$ for all t .

Thus

$$|\theta|_{H^{-a}}^2 = \sum_{k \leq K} k^{-2a} |\hat{\theta}_{\vec{k}}|^2 + \sum_{k > K} k^{-2a} |\hat{\theta}_{\vec{k}}|^2 \leq \sum_{k \leq K} k^{-2a} |\hat{\theta}_{\vec{k}}|^2 + K^{-2a} |\theta|_{L^2}^2. \quad (1.4.2)$$

For any $\varepsilon > 0$ choose $K(\varepsilon) \geq (2C/\varepsilon)^{1/2a}$

$$\implies k^{-2a} |\theta|_{L^2} \leq k^{-2a} C < \varepsilon/2 \quad (a > 0)$$

for $k \geq K(\varepsilon)$.

Then because for any finite K ,

$$\sum_{k \leq K} k^{-2a} |\hat{\theta}_{\vec{k}}(t)|^2 \rightarrow 0$$

as $t \rightarrow \infty$, there exists $N(\varepsilon) < \infty$ such that $\sum_{k \leq K(\varepsilon)} k^{-2a} |\hat{\theta}_{\vec{k}}(t)|^2 < \varepsilon/2$ for all $t > N(\varepsilon)$. From (1.4.2) we have that

$$|\theta|_{H^{-a}}^2 < \varepsilon$$

for all $t > T(\varepsilon)$, which ends the proof. $\square$

The above theorem clearly shows that $u(x, t)$ is mixing if there is a rapid decay of the H^{-a} norm. Also, these norms have the advantage that unlike variance they decay even in the absence of diffusion, and their decay corresponds to mixing in the sense of ergodic theory. This rigorous definition of a mixing in ergodic theory, is better suited for mathematical treatment of mixing.

The following chapters focus on the H^{-1} norm, which is related to the large-scale mixing measure and was studied by Thiffeault [Thi12]. The smaller the H^{-1} norm is, the more homogeneous the scalar field is.

1.4.1 Illustration I

For diffusion free case, Z.Lin, J.L Thiffeault, Doering [LTD11] test the optimal velocity $u_p = -\frac{\Delta^{-1}\mathbb{P}(\theta\nabla^{-1}\theta)}{\|\nabla^{-1}\mathbb{P}(\theta\nabla^{-1}\theta)\|_{L^2}}$ where $\mathbb{P}(v) = v - \nabla\Delta^{-1}(\nabla \cdot v)$ using $\theta_0 = \sin x$

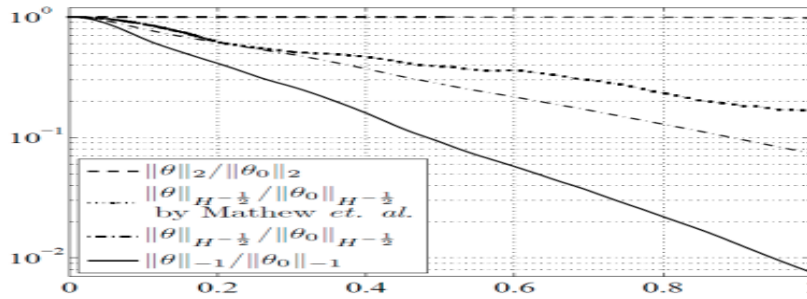


Figure 1.3: Comparison of the mix-norms for a flow optimized using the separate methods of optimal control and optimal instantaneous decay. Figure from Lin et al. [LTD11]. The solid line decays faster.

1.4.2 Illustration II

Also, for diffusion free case, using the optimal velocity $u_p = -\frac{\Delta^{-1}\mathbb{P}(\theta\nabla^{-1}\theta)}{\|\nabla^{-1}\mathbb{P}(\theta\nabla^{-1}\theta)\|_{L^2}}$ where $\mathbb{P}(v) = v - \nabla\Delta^{-1}(\nabla \cdot v)$ using $\theta_0 = \sin x$

1.5 Problem Statement

Advection-diffusion of a passive scalar field $\theta(x, t)$ by a smooth incompressible flow field $u(x, t)$ described by

$$\frac{\partial \theta}{\partial t} + u \cdot \nabla \theta = k \Delta \theta, \quad \theta(x, 0) = \theta_0(x) \quad (1.5.1)$$

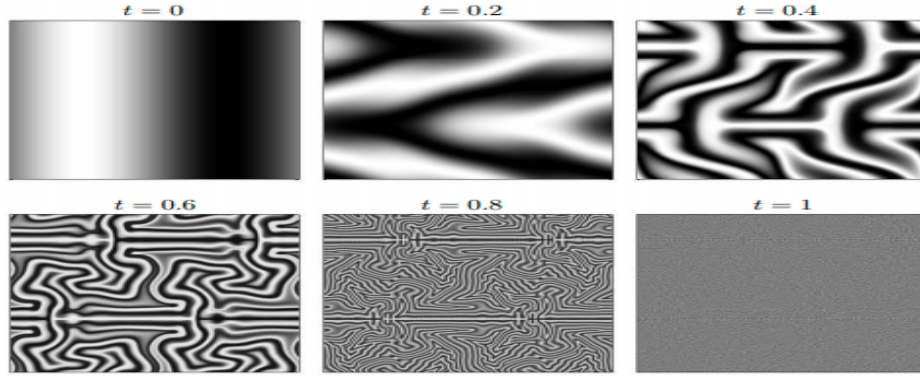


Figure 1.4: Evolution of the concentration field for the flow optimized in the case $a = 1$ as computed by Lin et al. Figure from Lin et al. [LTD11]

$$\nabla \cdot u = 0 \tag{1.5.2}$$

The stirring field u is periodic in the d -dimensional torus and L^∞ initial data θ_0 . The following cases are considered:

- $k = 0$
- $k \neq 0$
- $k \rightarrow 0$

1.6 Aim

The aim of this thesis is to examine “how well” tracers can be mixed under constraints on the advecting velocity field.

Chapter 2

Mixing without Diffusion

The results presented in this chapter follow the paper of Iyer, Kiselev, and Xu [IKX14], which establishes lower bounds on the mix norm for enstrophy-constrained incompressible flows, as well as the work of Lin, Thiffeault, and Doering [LTD11] on optimal stirring strategies. The exposition, calculations, and proofs below are drawn from these sources.

The central question is how well tracers can be mixed in the absence of diffusion. Following [IKX14], this chapter presents lower bounds on the mix norm and describes the optimal flow that maximizes the decay rate of this mix norm (when such flows exist). Consider the pure advection equation

$$\partial_t \theta + u \cdot \nabla \theta = 0 \quad (2.0.1)$$

with initial condition $\theta(x, 0) = \theta_0(x)$. The velocity field is assumed to be smooth and incompressible. The advecting velocity u and the initial data θ_0 are periodic in the d -dimensional torus $[0, L]^d$, as is θ .

It is also assumed that θ and θ_0 are spatially mean zero, that is:

$$\langle \theta(\cdot, t) \rangle \equiv \frac{1}{L^d} \int_{[0, L]^d} \theta(x, t) \, dx = 0 \quad (2.0.2)$$

The velocity fields u satisfy either the power constraint or the energy constraint.

Energy constraint

$$\int_{[0, L]^d} |u(x, t)|^2 \, dx = U^2 L^d \quad (2.0.3)$$

Enstrophy constraint

$$\int_{[0, L]^d} |\nabla u(x, t)|^2 \, dx = \frac{L^d}{\tau^2} \quad (2.0.4)$$

Given one of these constraints, the focus is on:

1. a lower bound for the mix norm.
2. the flow that maximizes the decay rate at every point in time.

2.1 Lower Bounds on the mix norm

In the absence of diffusion, incompressible flow conserves not only the variance of θ but also, by the weak maximum principle, the L^∞ norm: $\|\theta(\cdot, t)\|_{L^\infty} = \|\theta_0\|_{L^\infty}$ for all $t > 0$.

2.1.1 Energy constraint

$$\begin{aligned}
 \frac{d}{dt} \|\theta\|_{H^{-1}}^2 &= \frac{d}{dt} \int |\nabla^{-1} \theta|^2 \, dx \\
 &= \int \frac{\partial}{\partial t} |\nabla^{-1} \theta|^2 \, dx \\
 &= \int 2 \nabla^{-1} \theta \frac{\partial}{\partial t} \nabla^{-1} \theta \, dx \\
 &= \int 2 \nabla^{-1} \theta \nabla^{-1} \frac{\partial}{\partial t} \theta \, dx
 \end{aligned}$$

From (2.0.1), we have

$$\frac{d}{dt} \|\theta\|_{H^{-1}}^2 = \int 2 \nabla^{-1} \theta \nabla^{-1} (-u \cdot \nabla \theta) \, dx \tag{2.1.1}$$

Note that

$$\begin{aligned}
 \nabla \cdot (\theta u) &= \theta \nabla \cdot u + u \cdot \nabla \theta \\
 \nabla \cdot u = 0 &\implies \nabla \cdot (\theta u) = u \cdot \nabla \theta
 \end{aligned}$$

(2.1.1) $\implies$

$$\begin{aligned}
 \frac{d}{dt} \|\theta\|_{H^{-1}}^2 &= -2 \int \nabla^{-1} \theta \nabla^{-1} (u \cdot \nabla \theta) \, dx \\
 &= -2 \int \nabla^{-1} \theta \nabla^{-1} \nabla \cdot (\theta u) \, dx \\
 &= -2 \int \nabla^{-1} \theta \cdot (\theta u) \, dx \\
 &= -2 \int (\theta u) \cdot \nabla^{-1} \theta \, dx
 \end{aligned}$$

$$\frac{d}{dt} \|\theta\|_{H^{-1}}^2 = -2 \int \theta u \cdot \nabla (\Delta^{-1} \theta) \, dx \tag{2.1.2}$$

Using cauchy-schwarz inequality we have

$$\begin{aligned}
\frac{d}{dt} \|\theta\|_{H^{-1}}^2 &\geq -2 \|u\theta\|_{L^\infty} \|\nabla^{-1}\theta\|_{L^2} \\
&\geq -2 \|u\|_{L^2} \|\theta\|_{L^\infty} \|\nabla^{-1}\theta\|_{L^2} \\
&\geq -2 \|u\|_{L^2} \|\theta\|_{L^\infty} \|\theta\|_{H^{-1}} \\
&\geq -2UL^{\frac{d}{2}} \|\theta_0\|_{L^\infty} \|\theta\|_{H^{-1}}
\end{aligned}$$

$$2\|\theta\|_{H^{-1}} \frac{d}{dt} \|\theta\|_{H^{-1}} \geq -2UL^{\frac{d}{2}} \|\theta_0\|_{L^\infty} \|\theta\|_{H^{-1}}$$

Divide by $2\|\theta\|_{H^{-1}}$, we have

$$\frac{d}{dt} \|\theta\|_{H^{-1}} \geq -UL^{\frac{d}{2}} \|\theta_0\|_{L^\infty}$$

Integrating the equation from $s = 0$ and $s = t$

$$\begin{aligned}
\int_0^t d\|\theta\|_{H^{-1}} &\geq \int_0^t -UL^{\frac{d}{2}} \|\theta_0\|_{L^\infty} dt \\
\int_0^t d\|\theta(\cdot, t)\|_{H^{-1}} &\geq \int_0^t -UL^{\frac{d}{2}} \|\theta_0\|_{L^\infty} dt \\
\|\theta(\cdot, t)\|_{H^{-1}} - \|\theta(\cdot, 0)\|_{H^{-1}} &\geq -UL^{\frac{d}{2}} \|\theta_0\|_{L^\infty} t \\
\|\theta(\cdot, t)\|_{H^{-1}} &\geq \|\theta(\cdot, 0)\|_{H^{-1}} - UL^{\frac{d}{2}} \|\theta_0\|_{L^\infty} t
\end{aligned} \tag{2.1.3}$$

This lower bound clearly suggests that perfect mixing is possible in finite time and the time it takes, that is the minimal time for mixing t_{mix} , is given by the solution of the following equation:

$$\begin{aligned}
\|\theta(\cdot, 0)\|_{H^{-1}} - UL^{\frac{d}{2}} \|\theta_0\|_{L^\infty} t_{mix} &= 0 \\
\|\theta_0\|_{H^{-1}} &= UL^{\frac{d}{2}} \|\theta_0\|_{L^\infty} t_{mix} \\
t_{mix} &= \frac{1}{UL^{\frac{d}{2}}} \frac{\|\theta_0\|_{H^{-1}}}{\|\theta_0\|_{L^\infty}} = \frac{l_0}{2\pi U}
\end{aligned} \tag{2.1.4}$$

given that the length scale l_0 is

$$l_0 = \frac{2\pi \|\theta_0\|_{H^{-1}}}{L^{\frac{d}{2}} \|\theta_0\|_{L^\infty}} \tag{2.1.5}$$

By Poincare's inequality, we have that $l_0 \leq L$ where l_0 is basically the characteristic length of the initial nonhomogenous passive scalar. Of course, It is expected that the mixing time depends on the initial data as seen in (2.1.4).

2.1.2 Enstrophy constraint

From

$$\frac{d}{dt} \|\theta\|_{H^{-1}}^2 = -2 \int \nabla \Delta^{-1} \theta \cdot (\theta u) \, dx$$

Define the scalar field

$$\phi(x, t) = (\Delta^{-1} \theta)(x, t) = \sum_{k \neq 0} e^{ikx} k^{-2} \hat{\theta}_k(t) \quad (2.1.6)$$

where $\hat{\theta}_k(t)$ are the fourier coefficients, that is

$$\hat{\theta}_k(t) = \int e^{-ikx} \theta(x, t) \, dx$$

So,

$$\begin{aligned} \frac{d}{dt} \|\theta\|_{H^{-1}}^2 &= -2 \int \nabla \Delta^{-1} \theta \cdot (\theta u) \, dx \\ &= -2 \int \Delta^{-1} \theta \nabla u \Delta \Delta^{-1} \theta \, dx \\ &= -2 \int \phi \nabla u \Delta \phi \, dx \end{aligned}$$

Using Hölder inequality, we have

$$\begin{aligned} -2 \int \phi \nabla u \Delta \phi \, dx &\geq -2 \|\phi\|_{L^\infty} \|\nabla u \Delta \phi\|_{L^2} \\ \|\nabla u \Delta \phi\|_{L^2} &\leq \|\nabla u\|_{L^2} \|\Delta \phi\|_{L^2} \end{aligned}$$

this implies

$$\frac{d}{dt} \|\theta\|_{H^{-1}}^2 \geq -2 \|\phi\|_{L^\infty} \|\nabla u\|_{L^2} \|\Delta \phi\|_{L^2} dx$$

Using $\|\nabla u\|_{L^2}^2 = \frac{L^d}{\tau^2}$, then

$$\frac{d}{dt} \|\theta\|_{H^{-1}}^2 \geq -2 \|\phi\|_{L^\infty} \frac{L^{\frac{d}{2}}}{\tau} \|\theta_0\|_{L^2}$$

In three dimensions ($d=3$), the following interpolation estimate holds: For mean-zero functions on the 3-torus, there exists an $O(1)$ number C_3 such that

$$\|\phi\|_{L^\infty} \leq C_3 \|\nabla \phi\|_{L^2}^{\frac{1}{2}} \|\Delta \phi\|_{L^2}^{\frac{1}{2}} = C_3 \|\theta\|_{H^{-1}}^{\frac{1}{2}} \|\theta_0\|_{L^2}^{\frac{1}{2}} \quad (2.1.7)$$

$\Rightarrow$

$$\begin{aligned}
\frac{d}{dt} \|\theta\|_{H^{-1}}^2 &\geq -2 \|\phi\|_{L^\infty} \|\nabla u\|_{L^2} \|\Delta \phi\|_{L^2} \\
&= -2C_3 \|\nabla \phi\|_{L^2}^{\frac{1}{2}} \|\Delta \phi\|_{L^2}^{\frac{3}{2}} \|\nabla u\|_{L^2} \\
&= C_3 \|\theta\|_{H^{-1}}^{\frac{1}{2}} \|\theta_0\|_{L^2}^{\frac{3}{2}} \frac{L^{\frac{d}{2}}}{\tau} \\
\\
\frac{d}{dt} \|\theta\|_{H^{-1}}^2 &\geq -c_3 \|\theta\|_{H^{-1}}^{\frac{1}{2}} \|\theta_0\|_{L^2}^{\frac{3}{2}} \frac{L^{\frac{3}{2}}}{\tau} \\
2 \|\theta\|_{H^{-1}} \frac{d}{dt} \|\theta\|_{H^{-1}} &\geq -c_3 \|\theta\|_{H^{-1}}^{\frac{1}{2}} \|\theta_0\|_{L^2}^{\frac{3}{2}} \frac{L^{\frac{3}{2}}}{\tau} \\
\frac{d}{dt} \|\theta\|_{H^{-1}} &\geq -C_3 \|\theta\|_{H^{-1}}^{\frac{-1}{2}} \|\theta_0\|_{L^2}^{\frac{3}{2}} \frac{L^{\frac{3}{2}}}{\tau} \\
\frac{d \|\theta\|_{H^{-1}}}{\|\theta\|_{H^{-1}}^{\frac{-1}{2}}} &\geq -C_3 \|\theta_0\|_{L^2}^{\frac{3}{2}} \frac{L^{\frac{3}{2}}}{\tau} dt
\end{aligned} \tag{2.1.8}$$

Integrating with respect to t , we have

$$\begin{aligned}
\frac{2}{3} \|\theta\|_{H^{-1}}^{\frac{3}{2}} - \frac{2}{3} \|\theta_0\|_{H^{-1}}^{\frac{3}{2}} &\geq -C_3 \|\theta_0\|_{L^2}^{\frac{3}{2}} \frac{L^{\frac{3}{2}}}{\tau} \times t \\
\frac{2}{3} \|\theta\|_{H^{-1}}^{\frac{3}{2}} &\geq \frac{2}{3} \|\theta_0\|_{H^{-1}}^{\frac{3}{2}} - C_3 \|\theta_0\|_{L^2}^{\frac{3}{2}} \frac{L^{\frac{3}{2}}}{\tau} \times t \\
\|\theta\|_{H^{-1}} &\geq \|\theta_0\|_{H^{-1}} \left[1 - \frac{3C_3}{2\tau} \left(\frac{L \|\theta_0\|_{L^\infty}}{\|\theta_0\|_{H^{-1}}} \right)^{\frac{3}{2}} \times t \right]^{\frac{2}{3}}
\end{aligned} \tag{2.1.9}$$

provided the term in the square bracket is non-negative. *This also suggests that perfect mixing might be possible in finite time.* The mixing time is bounded below by

$$1 - \frac{3C_3}{2\tau} \left(\frac{L \|\theta_0\|_{L^\infty}}{\|\theta_0\|_{H^{-1}}} \right)^{\frac{3}{2}} \times t_{mix} = 0$$

$\Rightarrow$

$$t_{mix} = \tau \times \frac{2}{3C_3} \left(\frac{l_0}{2\pi L} \right)^{\frac{3}{2}} \tag{2.1.10}$$

where

$$l_0 = 2\pi \frac{\|\theta_0\|_{H^{-1}}}{\|\theta_0\|_{L^2}}$$

Using Poincare inequality, it is easy to see that $l_0 \leq L$. l_0 is another length scale describing the size of inhomogeneities in the initial data.

2.1.3 Uniformly bounded rate of strain constraint

In contrast to the finite-time mixing possibilities, consider the case where the rate of strain of the stirring velocity is uniformly bounded in space and time.

Given $\|\nabla u\| \leq \gamma < \infty$, and by Hölder's inequality, we have

$$\frac{d}{dt} \|\theta\|_{H^{-1}}^2 = 2 \int \nabla^{-1} \theta \cdot (\nabla u) \cdot \nabla^{-1} \theta dx \geq -2\gamma \|\theta\|_{H^{-1}}^2 \quad (2.1.11)$$

Using Gronwall's inequality, we have

$$\|\theta\|_{H^{-1}} \geq \|\theta_0\| e^{-\gamma t}$$

This suggests that perfect mixing is not possible in finite time but an exponential decay of the mix norm.

2.1.4 Optimal striring

Multiplying (2.0.1) by $-\Delta^{-1}\theta$, and integrating over the spatial domain, and by parts we have

$$\frac{d}{dt} \|\nabla^{-1} \theta\|_{L^2}^2 = \frac{d}{dt} \|\theta\|_{H^{-1}}^2 = -2 \int \theta u \cdot \nabla(\Delta^{-1} \theta) dx \quad (2.1.12)$$

Define $\phi(x, t) = \Delta^{-1}\theta(x, t)$ and write

$$\frac{d}{dt} \|\nabla^{-1} \theta\|_{L^2}^2 = -2 \int u \cdot (\theta \nabla \phi) dx = -2 \int u \cdot \mathbb{P}(\theta \nabla \phi) dx \quad (2.1.13)$$

where $\mathbb{P}$ is the projector onto divergence-free fields defined by

$$\mathbb{P}(v) = v \cdot \nabla \Delta^{-1}(\nabla \cdot v).$$

Now suppose an enstrophy constraint,

$$\int_{[0,L]^d} |\nabla u(x, t)|^2 dx = U^2 L^d$$

$$U = \frac{1}{L^{\frac{d}{2}}} \left(\int_{[0,L]^d} |\nabla u(x, t)|^2 dx \right)^{\frac{1}{2}} = \frac{1}{L^{\frac{d}{2}}} \|\nabla u\|_{L^2}$$

In particular, the Cauchy-Schwarz inequality gives

$$\begin{aligned} \int u \cdot \mathbb{P}(\theta \nabla \phi) dx &\leq \|u\|_{L^2} \|\mathbb{P}(\theta \nabla \phi)\|_{L^2} \\ &\leq U L^{\frac{d}{2}} \|\mathbb{P}(\theta \nabla \phi)\|_{L^2} \end{aligned}$$

The optimal u_e saturating this inequality satisfies

$$\int u \cdot \mathbb{P}(\theta \nabla \phi) dx = UL^{\frac{d}{2}} \|\mathbb{P}(\theta \nabla \phi)\|_{L^2}.$$

Solving the equation above gives

$$\begin{aligned} u_e &= \frac{U \mathbb{P}(\theta \nabla \phi)}{\frac{1}{L^{\frac{d}{2}}} \left(\int_{\Omega} |\mathbb{P}(\theta \nabla \phi)|^2 dx \right)^{\frac{1}{2}}} \\ u_e &= U \frac{\mathbb{P}(\theta \nabla \phi)}{\langle |\mathbb{P}(\theta \nabla \phi)|^2 \rangle^{\frac{1}{2}}} \end{aligned} \quad (2.1.14)$$

With fixed enstrophy constraint $\int_{[0,L]^d} |\nabla u(x,t)|^2 dx = \frac{L^d}{\tau^2}$, the optimal mixer maximizing the rate of decay of the mix- norm H^{-1} is

$$u_p = \frac{1}{\tau} \frac{-\Delta^{-1} \mathbb{P}(\theta \nabla \phi)}{\langle |\nabla^{-1} \mathbb{P}(\theta \nabla \phi)|^2 \rangle^{\frac{1}{2}}} \quad (2.1.15)$$

provided the norm in denominator does not vanish.

2.2 Lower Bound on mix-norm for enstrophy constrained flow

The numerical simulation by Lin and Charles [LTD11] suggests that finite time perfect mixing by an enstrophy constrained incompressible is impossible but a robust exponential decay of the mix norm. The rigorous analysis suggests that it may be possible. In the next section, this discrepancy between analysis and simulation is addressed. The following theorem settles this affirmatively.

Theorem 2 ([IKX14, Theorem 1.1]). *Let u be a smooth time independent incompressible periodic vector field on the d -dimensional torus, and let θ solve (2.0.1) with periodic boundary conditions and L_1 initial data θ_0 . For any $p > 1$ and $\lambda \in (0, 1)$ there exists a length scale $r_0 = r_0(\theta_0, \lambda)$, an explicit constant $\varepsilon_0 = \varepsilon_0(\lambda, d)$, and a constant $c = c(d, p)$ such that*

$$\|\theta(t)\|_{H^{-1}} \geq \varepsilon_0 r_0^{\frac{d}{2}+1} \|\theta_0\|_{L^\infty} \exp \left(\frac{-c}{m(A_\lambda)^{\frac{1}{p}}} \int_0^t \|\nabla u(s)\|_{L^p} ds \right) \quad (2.2.1)$$

Here A_λ is the super level set $\{\theta_0 > \lambda \|\theta_0\|_{L^\infty}\}$

It is straightforward to see from the theorem that if the instantaneous enstrophy constraint $\|\nabla u(s)\|_{L^2} \leq F$ is enforced, then $\|\theta(t)\|_{H^{-1}}$ decays almost exponentially in time.

From (2.2.1), it follows that passive scalars cannot be mixed perfectly in finite time.

2.2.1 δ - mixed

Towards the proof of Theorem 2, the following definition and lemmas are needed. The numerical simulations presented later indicate an exponential lower bound with a decay rate in good qualitative agreement with (2.2.1).

Definition 2.2.1 ([IKX14, Definition 2.1]). Let $k \in (0, \frac{1}{2})$ be fixed. For $\delta > 0$, we say a set $A \subset \mathbb{T}^d$ is δ -semi-mixed if

$$\frac{m(A \cap B(x, \delta))}{m(B(x, \delta))} \leq 1 - k \quad \forall x \in \mathbb{T}^d \quad (2.2.2)$$

Also, A^c is also δ -semi-mixed, then we say A is δ -mixed.

Relation Between δ -semi-mixed and negative Sobolev norms

Lemma 2.2.2 ([IKX14, Lemma 2.2]). Let $\lambda \in (0, 1]$ and $\theta \in L^\infty(\mathbb{T}^d)$. If for any integer $n > 0$, $k \in (0, \frac{\lambda}{1+\lambda})$ there exists an explicit constant $c_0 = c_0(d, k, \lambda, n)$ such that

$$\|\theta\|_{H^{-n}} \leq \frac{\|\theta\|_{L^\infty} \delta^{n+\frac{d}{2}}}{c_0} \quad (2.2.3)$$

$\implies A_\lambda$ is δ -semi-mixed.

Here, A_λ is the super level set defined by $A_\lambda = \{\theta > \lambda \|\theta\|_{L^\infty}\}$

Proof. Assume by contradiction that A_λ is not δ -semi-mixed. Then by definition 2.2.1, there exists $x \in \mathbb{T}^d$ such that

$$\frac{m(A_\lambda \cap B(x, \delta))}{m(B(x, \delta))} \geq (1 - \kappa)$$

$\implies$

$$m(A_\lambda \cap B(x, \delta)) \geq (1 - \kappa)m(B(x, \delta)) = (1 - \kappa)\pi(d)\delta^d. \quad (2.2.4)$$

Where $\pi(d)$ is the volume of d -dimensional unit ball. Using duality

$$|\theta|_{H^{-n}} = \sup_{f \in H^n} \frac{1}{|f|_{H^n}} \left| \int_{\mathbb{T}^d} \theta(x) f(x) dx \right|. \quad (2.2.5)$$

Let $f \in H^n$ be a function which is identically equal to 1 in $B(x, \delta)$, and which vanishes outside $B(x, (1 + \varepsilon)\delta)$ for some small $\varepsilon > 0$. A simple calculation shows that

$$|f|_{H^n} \leq c_1(d) \cdot \varepsilon^{-n+\frac{1}{2}} \cdot \delta^{-n+\frac{d}{2}},$$

for some (explicit) constant c_1 depending only on the dimension.

On the other hand using (2.2.4) gives

$$\int_{\mathbb{T}^d} \theta(x) f(x) dx \geq \pi(d) |\theta|_{L^\infty} \delta^d ((1 - \kappa)\lambda - \kappa - c_2(d)\varepsilon), \quad (2.2.6)$$

for some (explicit) dimensional constant $c_2(d)$. Choosing $\varepsilon = \frac{\lambda-(1+\lambda)\kappa}{2c_2(d)}$ and using (2.2.5) we obtain

$$|\theta|_{H^{-n}} \geq \frac{|\theta|_{L^\infty} \delta^{n+\frac{d}{2}}}{c_0(d, \kappa, \lambda, n)}$$

as desired. $\square$

2.2.2 Lemma and Proof of Lower Bound Theorem

Lemma 2.2.3 ([IKX14, Lemma 2.3]). *Let ψ be the flow map of an incompressible vector field u . Let $A \subset \mathbb{T}^d$ be any measurable set and $p > 1$. There exist constants $r_0 = r_0(A)$ and $a = a(d, k, p) > 0$, such that if for some $\delta < \frac{r_0}{2}$ and $\tau > 0$ and the set $\psi_T(A)$ is δ -semi-mixed, then*

$$\int_0^T \|\nabla u(\cdot, t)\|_{L^p} dt \geq \frac{m(A)^{\frac{1}{p}}}{a} \left| \ln \left(\frac{2\delta}{r_0} \right) \right| \quad (2.2.7)$$

Proof. of Theorem 2. Replacing θ with $\frac{\theta}{\|\theta\|_{L^\infty}}$, without the loss of generality, assume $\|\theta_0\|_{L^\infty} = 1$. Fix $0 < \lambda \leq 1$ and $k \in (0, \frac{\lambda}{1+\lambda})$.

Let a be a constant from Lemma 2.2.3, and c_0 be a constant from Lemma 2.2.2 with $n = 1$ (i.e H^{-1})

Choose

$$\delta = (c_0 \|\theta(t)\|_{H^{-1}})^{\frac{2}{d+2}}$$

Then

$$\|\theta(t)\|_{H^{-1}} = \frac{\delta^{\frac{d}{2}+1}}{c_0} \quad (2.2.8)$$

Using Lemma 2.2.2, it implies the superlevel set $\{\theta(t) > \lambda\}$ is δ -semi-mixed.

Let $\{\theta(t) > \lambda\} = \psi(A_\lambda)$ where ψ is the flow of the vector field u .

Using Lemma 2.2.3 implies

$$\begin{aligned} \int_0^T \|\nabla u(t)\|_{L^p} dt &\geq \frac{m(A)^{\frac{1}{p}}}{a} \left| \ln \left(\frac{2\delta}{r_0} \right) \right| \\ \frac{r_0}{2} \exp \left(\frac{-a}{m(A)^{\frac{1}{p}}} \int_0^T \|\nabla u(t)\|_{L^p} dt \right) &\leq \delta \end{aligned} \quad (2.2.9)$$

From (2.2.8), we have

$$\|\theta(t)\|_{H^{-1}} = \frac{\delta^{\frac{d}{2}+1}}{c_0} \geq \frac{r_0^{\frac{d}{2}+1}}{c_0 2^{\frac{d}{2}+1}} \exp \left(\frac{-da}{m(A_\lambda)^{\frac{1}{p}}} \int_0^T \|\nabla u(t)\|_{L^p} dt \right)$$

Pick $\varepsilon_0 = \frac{1}{c_0 2^{\frac{d}{2}+1}}$, $c = -da$ ends the proof. $\square$

2.3 Universal Decay Rates

This section examines what happens when the advecting velocity is zero at the boundary, i.e., no-slip conditions, following [IKX14].

Proposition 2.3.1 ([IKX14, Proposition 3.1]). *Let $I = (0, \ell)^d$ be a cube in $\mathbb{R}^d$. Suppose that there exist $k \in \mathbb{R}$, $q \in [1, \infty]$ and $c_0 > 0$ such that*

$$|\theta(t)|_{H^{-1}} \geq B(\theta_0) \exp\left(\frac{-c_0}{\ell^{d/p}} \int_0^t |\nabla u|_{L^p}\right) \quad (2.3.1)$$

for some $p \in [1, d]$, all incompressible u which vanish on ∂I , and all initial data $\theta_0 \in C_c^\infty(I)$. Assume that there exists $\gamma \in \mathbb{R}$ such that the pre-factor $B(\theta_0)$ satisfies

$$B(A\theta_0) = AB(\theta_0) \quad \text{and} \quad B(\theta_{0,a}) = a^{-\gamma} B(\theta_0),$$

where $\theta_{0,a}(x) = \theta_0(x/a)$ for $x \in (0, a)^d$ and $\theta_{0,a}(x) = 0$ otherwise.

Then, for any mean zero $\theta_0 \in C_c^\infty(I)$ and any smooth velocity field u such that

$$\limsup_{t \rightarrow \infty} \frac{1}{t} \int_0^t |\nabla u|_{L^p} < \infty, \quad \nabla \cdot u = 0, \quad \text{and} \quad u = 0 \text{ on } \partial I,$$

the decay of $|\theta(t)|_{H^{-1}}$ as $t \rightarrow \infty$ is bounded below by an algebraic function of time.

Proof. Without loss of generality, assume $\ell = 1$. Let $\theta_0 \in C_c^\infty$ be zero mean initial data and let $h(t) = |\theta(t)|_{H^{-1}}$.

Aim: To show that h decays algebraically with respect to t as t goes large.

Let $a > 0$, and

$$I_a = (0, a)^d, \quad \eta(x, t) = \chi_{I_a} \times \theta(x/a, t/a), \quad v(x, t) = \chi_{I_a} \times u(x/a, t/a).$$

Then (η, v) is a solution of (2.0.1) on the *unscaled* domain I .

Because $\theta_0 \in C_c^\infty(I)$ and $u = 0$ on ∂I , we see $\theta(t)$ remains compactly supported in I_a for all $t > 0$. We assumed that,

$$B(\eta_0) = B(\chi_{I_a} \times \theta_0(\frac{x}{a})) = a^{-\gamma} B(\theta_0) \quad \text{and} \quad |\nabla v(t)|_{L^p} = a^{\frac{d}{p}-1} |\nabla u(t/a)|_{L^p}.$$

and using equation (2.3.1), the first equality by duality and scaling gives

$$\begin{aligned} a^{d/2+1} h(t/a) &= |\eta(t)|_{H^{-1}} \geq B(\eta_0) \exp\left(-c_0 \int_0^t |\nabla v|_{L^p}\right) \\ &= a^{-\gamma} B(\theta_0) \exp\left(-c_0 a^{\frac{d}{p}-1} \int_0^t |\nabla u(s/a)|_{L^p} ds\right). \end{aligned}$$

Hence, taking $t' = t/a$ gives

$$h(t') \geq a^{-N} B(\theta_0) \exp \left(-c_0 a^{d/p} \int_0^{t'} |\nabla u(s)|_{L^p} ds \right),$$

where $N = d/2 + 1 + \gamma$. This lower bound is true $\forall a > 0$. Let

$$f(a) = a^{-N} B(\theta_0) \exp \left(-c_0 a^{d/p} U \right),$$

where $U = \int_0^{t'} |\nabla u(s)|_{L^p} ds$

Maximizing $f(a)$ in a (and changing t' to t), we have

$$f'(a) = -N a^{-N-1} B(\theta_0) \exp \left(-c_0 a^{d/p} U \right) + a^{-N} B(\theta_0) \exp \left(-c_0 a^{d/p} U \right) - c_0 \frac{d}{p} a^{d/p-1} U$$

$$f'(a) = 0$$

$$\implies -N a^{-1} - c_0 \frac{d}{p} a^{\frac{d}{p}-1} U = 0$$

$$\implies a = \left(\frac{-N}{c_0 U^{\frac{d}{p}}} \right)^{\frac{p}{d}}$$

Substituting a in $f(a)$, we arrive at an algebraic lower bound

$$h(t) \geq C \left(\frac{N}{\int_0^t \|\nabla u\|_{L^p} ds} \right)^{-pN/d}.$$

$$|\theta(t)|_{H^{-1}} \geq C \left(\frac{N}{\int_0^t \|\nabla u\|_{L^p} ds} \right)^{-pN/d}.$$

□

2.4 Numerical Discussion

This section presents the author's own Python pseudo-spectral simulation of the optimal mixing problem, written as a port of Gautam Iyer's MATLAB code that accompanies [IKX14, LTD11]. The implementation uses $N = 64$ spectral modes per spatial direction on the unit torus $\mathbb{T}^2 = [0, 1]^2$, with Dormand–Prince RK45 time integration (tolerances `rtol` = 10^{-6} , `atol` = 10^{-8}). Under incompressible advection all L^p norms of θ are conserved; the simulation is terminated automatically once the L^2 , L^4 and L^8 norms drift by more than 10^{-3} from their initial values, indicating that the grid can no longer resolve the fine-scale structures generated by stirring.

Optimal velocity and initial data

The Lin–Thiffeault–Doering optimal velocity [LTD11], which minimises $\frac{d}{dt}\|\theta\|_{H^{-1}}^2$ instantaneously subject to the enstrophy constraint $\|\nabla u\|_{L^2} \leq F$, is

$$u = F \frac{v}{\|\nabla v\|_{L^2}}, \quad v = -\Delta^{-1}\mathbb{P}(\theta \nabla \Delta^{-1}\theta),$$

where $\mathbb{P}(g) = g - \nabla \Delta^{-1}(\nabla \cdot g)$ is the Leray projection onto divergence-free fields. The constraint force $F = 1$ is used throughout.

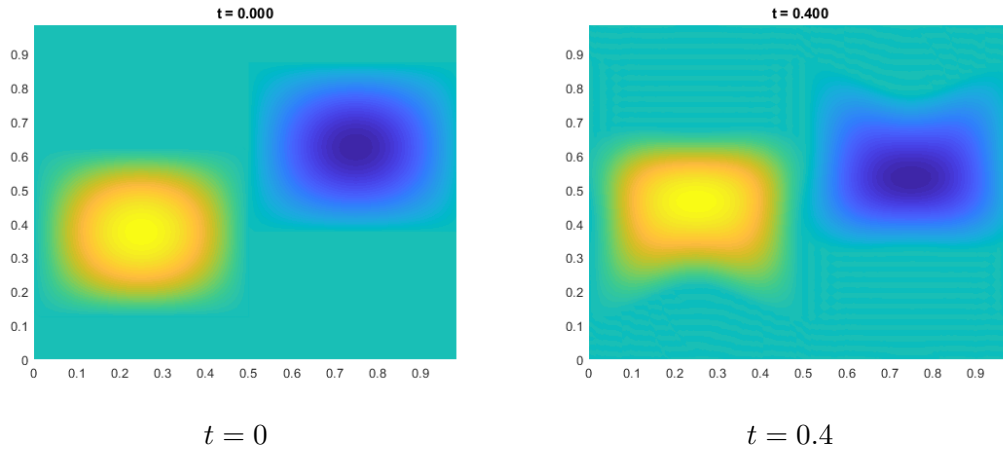
The initial data is parametrised by a support-size parameter $b \in (0, 1)$:

$$\theta_0(x, y) = \begin{cases} \sin\left(\frac{2\pi x}{b}\right) \sin\left(\frac{2\pi(y + \frac{b}{8})}{b}\right) & \text{for } 0 < x < \frac{b}{2}, \quad \frac{-b}{8} < y < \frac{b}{2} - \frac{b}{8}, \\ \sin\left(\frac{2\pi x}{b}\right) \sin\left(\frac{2\pi(y - \frac{b}{8})}{b}\right) & \text{for } \frac{b}{2} < x < b, \quad \frac{b}{8} < y < \frac{b}{2} + \frac{b}{8}, \\ 0 & \text{otherwise,} \end{cases}$$

normalised so that $\|\theta_0\|_{L^2} = 1$. Since $m(\text{supp}(\theta_0)) = O(b^2)$, Theorem 2 predicts that the exponential decay rate $\lambda(b)$ should scale linearly in b .

Snapshots and mix-norm decay

The figures below show six snapshots of the scalar field $\theta(x, y, t)$ for $b = 1$, sampled from $t = 0$ to the stopping time. The LTD velocity progressively stretches and folds the scalar into finer filaments, consistent with an exponential decrease in the H^{-1} norm.



Dependence on support size b

Eight values of b are simulated and the exponential decay rate $\lambda(b)$ is extracted by fitting a line to $\log \|\theta\|_{H^{-1}}$ over the final two-thirds of each trajectory (discarding the initial transient). The results are summarised in Table 2.1.

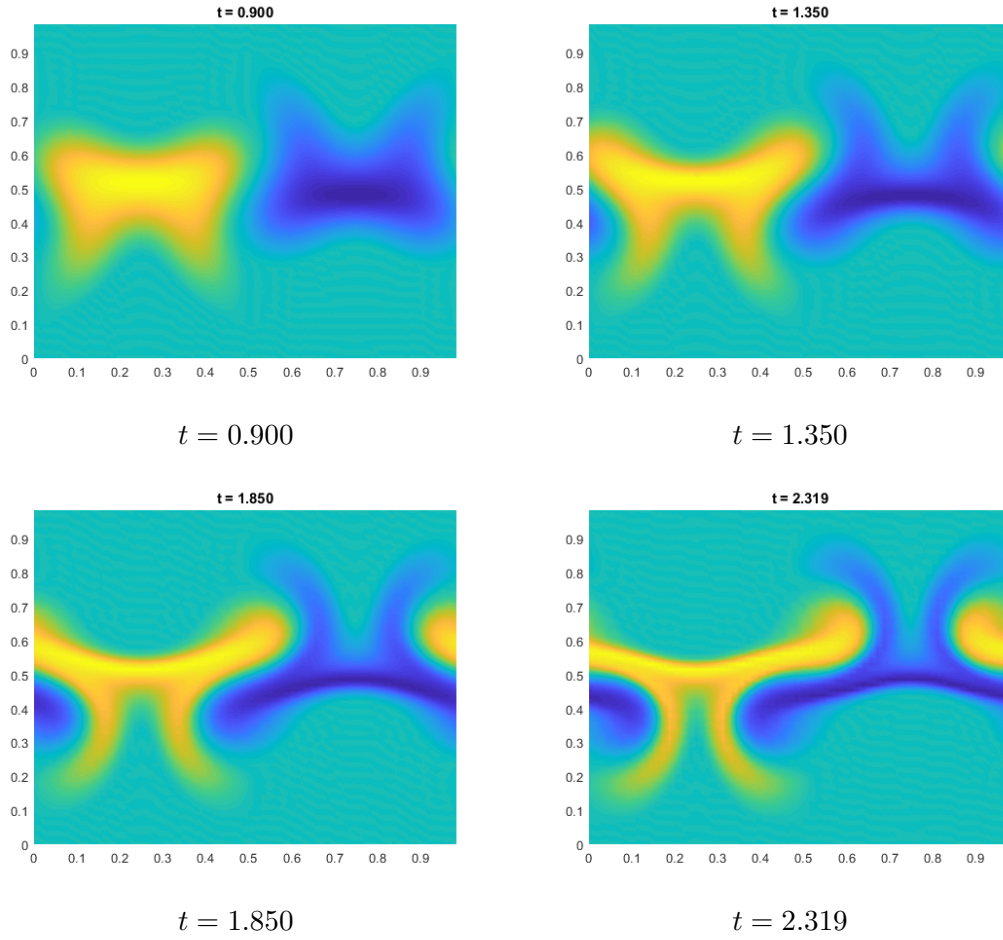


Figure 2.3: Snapshots of the passive scalar $\theta(x, y, t)$ at six times for $b = 1$, $N = 64$, $F = 1$. The LTD velocity continuously sharpens scalar gradients into progressively finer structures.

Table 2.1: Fitted exponential decay rates λ and mixing timescales $\tau = -1/\lambda$ for eight support-size values b . ($N = 64$, $F = 1$.)

b	t_{stop}	λ	$\tau = -1/\lambda$
0.500	1.30	-0.440	2.27
0.563	1.60	-0.386	2.59
0.625	1.95	-0.334	2.99
0.688	2.30	-0.286	3.49
0.750	2.70	-0.244	4.11
0.813	3.00	-0.207	4.82
0.875	3.25	-0.174	5.74
0.938	3.20	-0.142	7.03

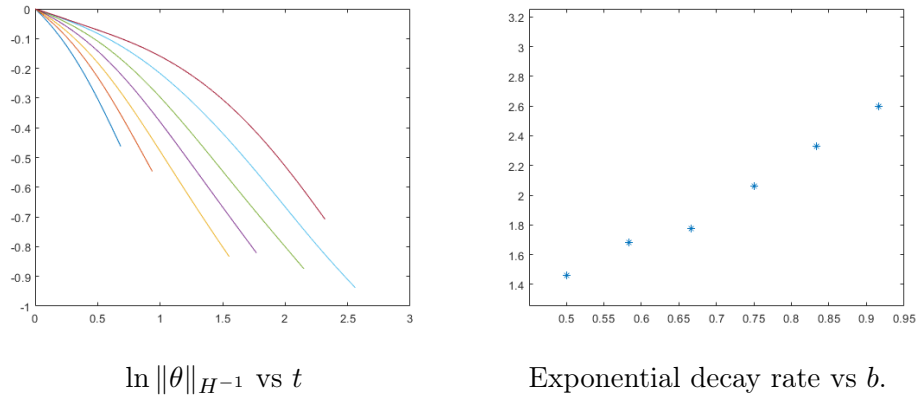


Figure 2.4: Left: $\log$ of the H^{-1} mix norm vs. time for $b = 1$; the near-linear decay confirms approximate exponential mixing. Right: fitted exponential decay rate $\lambda(b)$ for eight values of b .

The decay rate λ grows as b decreases: scalars with narrower initial support mix faster. A power-law fit to $\log |\lambda|$ vs $\log b$ gives $\lambda \sim b^{-1.78}$, compared with the theoretically predicted b^{-1} scaling of [LTD11]. This deviation is consistent with the known gap between the greedy LTD strategy and the global optimum [IKX14]; higher resolution (larger N) would be needed to access the true asymptotic regime. Qualitatively, Figure 2.4b confirms the linear relationship between the decay rate and b predicted by Theorem 2, since $m(\text{supp}(\theta_0)) = O(b^2)$.

2.5 Conclusion

1. In the absence of diffusion, finite-time perfect mixing might
 - be possible by an energy constrained flows.
 - be impossible by an enstrophy constrained flows.
2. Finally, Proposition 2.3.1 proves that the exponential decay rate in mixing with an enstrophy constrained flow would lead to an algebraic in time lower bound if the stirring velocity field vanishes at the boundary and the initial data is zero on the boundary.

Chapter 3

Mixing with Diffusion

The results in this chapter follow the analysis of Lunasin, Lin, Novikov, Mazzucato, and Doering [LLN⁺12] on optimal mixing and stirring under energy and enstrophy constraints in the presence of diffusion. The lower bounds and calculations presented below are drawn from that work.

Consider the advection of a passive scalar field $\theta(x, t)$ by a smooth incompressible flow field $u(x, t)$ described by

$$\frac{\partial \theta}{\partial t} + u \cdot \nabla \theta = \frac{1}{Pe} \Delta \theta, \quad \theta(x, 0) = \theta_0(x) \quad (3.0.1)$$

$$\nabla \cdot u = 0 \quad (3.0.2)$$

The stirring field u is periodic in the d -dimensional torus and L^∞ initial data θ_0 . Given two constraints

- *Energy constraint.* $\|u(t)\|_{L^2} = UL^{\frac{d}{2}}$
- *Enstrophy constraint.* $\|\nabla u(t)\|_{L^2} = \Gamma L^{\frac{d}{2}}$

3.1 Aim

The central questions addressed in this chapter, following [LLN⁺12], are:

- Does the presence of diffusion aid mixing under constraints on the velocity field?
- What is the optimal mixing rate in the presence of diffusion under energy and enstrophy constraints?

3.2 Lower bounds

Rate of mixing:

Following [LLN⁺12], the rate of mixing is defined using the H^{-1} norm as follows:

$$r(t) = -\frac{\frac{d}{dt} \|\nabla^{-1}\theta\|_{L^2}}{\|\nabla^{-1}\theta\|_{L^2}} \quad (3.2.1)$$

and

Filamentation length measure as:

$$\lambda(t) = 2\pi \frac{\|\nabla^{-1}\theta(\cdot, t)\|_{L^2}}{\|\theta(\cdot, t)\|_{L^2}} \quad (3.2.2)$$

3.2.1 Enstrophy constrained $\|\nabla u\|_{L^\infty} = 1$

$$\begin{aligned} \lambda(t) &= 2\pi \frac{\|\nabla^{-1}\theta\|_{L^2}}{\|\theta\|_{L^2}} \\ \partial_t \theta + u \cdot \nabla \theta &= \frac{1}{Pe} \nabla \theta \\ \lambda^2 &= 4\pi^2 \frac{\|\nabla^{-1}\theta\|_{L^2}^2}{\|\theta\|_{L^2}^2} \\ \frac{d\lambda^2}{dt} &= 4\pi^2 \frac{\|\theta\|_{L^2}^2 \frac{d}{dt} \|\nabla^{-1}\theta\|_{L^2}^2 - \|\nabla^{-1}\theta\|_{L^2}^2 \frac{d}{dt} \|\theta\|_{L^2}^2}{\|\theta\|_{L^2}^4} \end{aligned} \quad (3.2.3)$$

Multiply (3.0.1) by θ ,

$$\begin{aligned} \theta \partial_t \theta + \theta u \cdot \nabla \theta &= \frac{1}{Pe} \theta \Delta \theta \\ \frac{1}{2} \partial_t \theta^2 + \theta u \cdot \nabla \theta &= \frac{1}{Pe} \theta \nabla \cdot \nabla \theta \\ \frac{1}{2} \partial_t \theta^2 + \theta u \cdot \nabla \theta &= \frac{1}{Pe} (\nabla \cdot (\theta \nabla \theta) - \nabla \theta \cdot \nabla \theta) \\ \frac{1}{2} \partial_t \theta^2 - \frac{1}{Pe} |\nabla \theta|^2 &= \frac{1}{Pe} \nabla \cdot (\theta \nabla \theta) - \theta u \cdot \nabla \theta \end{aligned}$$

and integrate over the domain D

$$\frac{1}{2} \int_D \partial_t \theta^2 \, d^d x - \int_D \frac{1}{Pe} |\nabla \theta|^2 \, d^d x = \int_D \frac{1}{Pe} \nabla \cdot (\theta \nabla \theta) \, d^d x - \int_D \theta u \cdot \nabla \theta \, d^d x$$

Applying the boundary conditions, we have

$$\frac{d}{dt} \|\theta\|_{L^2}^2 = \frac{-2}{Pe} \|\nabla \theta\|_{L^2}^2. \quad (3.2.4)$$

Also,

$$\frac{d\lambda^2}{dt} = 2 \int \nabla^{-1} \theta \cdot (\nabla u) \cdot \nabla^{-1} \theta \, dx \quad (3.2.5)$$

Using (3.2.4) and (3.2.5) in (3.2.3), this implies

$$\frac{1}{4\pi^2} \frac{d\lambda^2}{dt} = \frac{2}{Pe} \left[\frac{\|\nabla\theta\|_{L^2}^2 \|\nabla^{-1}\theta\|_{L^2}^2}{\|\theta\|_{L^2}^4} \right] + \frac{2 \int_D \nabla^{-1}\theta \cdot \nabla u \cdot \nabla^{-1}\theta \, d^d x}{\|\theta\|_{L^2}^2}$$

It is easy to see that using holder's inequality, we deduce

$$\int_D \nabla^{-1}\theta \cdot \nabla u \cdot \nabla^{-1}\theta \, d^d x \leq \|\nabla u\|_{L^\infty} \int_D |\nabla^{-1}\theta|^2 \, d^d x = \|\nabla^{-1}\theta\|_{L^2}^2$$

where we used that $\|\nabla u\|_{L^\infty} = 1$.

We obtained

$$\frac{d\lambda^2}{dt} \geq \frac{2}{Pe} \left[\frac{\|\nabla\theta\|_{L^2}^2 \|\nabla^{-1}\theta\|_{L^2}^2}{\|\theta\|_{L^2}^4} \right] - 2\lambda^2 \quad (3.2.6)$$

Using one interpolation inequality, it is clear that

$$\frac{\|\nabla\theta\|_{L^2}^2 \|\nabla^{-1}\theta\|_{L^2}^2}{\|\theta\|_{L^2}^4} \geq 0,$$

Then (3.2.6) implies

$$\frac{d\lambda^2}{dt} \geq -2\lambda^2$$

By Gronwall's inequality, we have:

$$\lambda(t) > \lambda(0)e^{-t}, \quad (3.2.7)$$

$$\frac{\|\nabla^{-1}\theta\|_{L^2}}{\|\theta\|_{L^2}} \geq \frac{\|\nabla^{-1}\theta_0\|_{L^2}}{\|\theta_0\|_{L^2}} e^{-t}$$

This shows that perfect mixing is not possible in finite time for L^∞ bounded rate of strain flows.

Consider now

$$\frac{d}{dt} \left(\frac{\|\nabla\theta\|_{L^2}^2}{\|\theta\|_{L^2}^2} \right) = \frac{\|\theta\|_{L^2}^2 \frac{d}{dt} \|\nabla\theta\|_{L^2}^2 - \|\nabla\theta\|_{L^2}^2 \frac{d}{dt} \|\theta\|_{L^2}^2}{\|\theta\|_{L^2}^4}$$

Note that

$$\begin{aligned} \frac{d}{dt} \|\nabla\theta\|_{L^2}^2 &= -2 \int_D \partial_t \theta \Delta \theta \, d^d x \\ &= -2 \int_D \left(\frac{1}{Pe} \Delta \theta - u \cdot \nabla \theta \right) \Delta \theta \, d^d x \\ &= \frac{-2}{Pe} \int_D |\Delta \theta|^2 \, d^d x + 2 \int_D u \cdot \nabla \theta \cdot \Delta \theta \, d^d x \\ &= \frac{-2}{Pe} \int_D |\Delta \theta|^2 \, d^d x + 2 \int_D \nabla \theta \cdot \nabla u \cdot \nabla \theta \, d^d x \end{aligned}$$

$$\begin{aligned}
\frac{d}{dt} \left(\frac{||\nabla\theta||_{L^2}^2}{||\theta||_{L^2}^2} \right) &= \frac{||\theta||_{L^2}^2 \left(\frac{-2}{Pe} ||\Delta\theta||_{L^2}^2 - 2 \int_D \nabla\theta \cdot \nabla u \cdot \nabla\theta \, d^d x \right)}{||\theta||_{L^2}^4} - \frac{||\nabla\theta||_{L^2}^2 \left(\frac{-2}{Pe} ||\nabla\theta||_{L^2}^2 \right)}{||\theta||_{L^2}^4} \\
&= \frac{-2}{Pe} \left(\frac{||\Delta\theta||_{L^2}^2}{||\theta||_{L^2}^2} - \frac{||\nabla\theta||_{L^2}^4}{||\theta||_{L^2}^4} \right) - 2 \frac{\int_D \nabla\theta \cdot \nabla u \cdot \nabla\theta \, d^d x}{||\theta||_{L^2}^2}
\end{aligned}$$

This yields

$$\frac{d}{dt} \left(\frac{||\nabla\theta||_{L^2}^2}{||\theta||_{L^2}^2} \right) \leq \frac{2||\nabla\theta||_{L^2}^2}{||\theta||_{L^2}^2}$$

By Gronwall estimate, we have

$$\begin{aligned}
\frac{||\nabla\theta||_{L^2}^2}{||\theta||_{L^2}^2} &\leq \frac{||\nabla\theta_0||_{L^2}^2}{||\theta_0||_{L^2}^2} e^{2t} \\
\Rightarrow \\
||\nabla\theta||_{L^2}^2 &\leq \frac{||\nabla\theta_0||_{L^2}^2}{||\theta_0||_{L^2}^2} e^{2t} ||\theta||_{L^2}^2
\end{aligned} \tag{3.2.8}$$

From (3.2.4), and using (3.2.8)

$$\begin{aligned}
\frac{d}{dt} ||\theta||_{L^2}^2 &= \frac{-2}{Pe} ||\nabla\theta||_{L^2}^2 \geq \frac{-2}{Pe} ||\theta||_{L^2}^2 \frac{||\nabla\theta_0||_{L^2}^2}{||\theta_0||_{L^2}^2} e^{2t} \\
\frac{\frac{d}{dt} ||\theta||_{L^2}^2}{||\theta||_{L^2}^2} &\geq \frac{-2}{Pe} \frac{||\nabla\theta_0||_{L^2}^2}{||\theta_0||_{L^2}^2} e^{2t} \\
\frac{d||\theta||_{L^2}^2}{||\theta||_{L^2}^2} &\geq \frac{-2}{Pe} \frac{||\nabla\theta_0||_{L^2}^2}{||\theta_0||_{L^2}^2} e^{2t} dt
\end{aligned}$$

Integrating from $s = 0$ to $s = t$

$$\begin{aligned}
\int_0^t \frac{d||\theta||_{L^2}^2}{||\theta||_{L^2}^2} &\geq \int_0^t \frac{-2}{Pe} \frac{||\nabla\theta_0||_{L^2}^2}{||\theta_0||_{L^2}^2} e^{2t} \, dt \\
\Rightarrow \\
||\theta||_{L^2} &\geq ||\theta_0||_{L^2} \exp \left[\frac{-1}{2Pe} \frac{||\nabla\theta_0||_{L^2}^2}{||\theta_0||_{L^2}^2} (e^{2t} - 1) \right]
\end{aligned} \tag{3.2.9}$$

From (3.2.7), we have

$$\begin{aligned}
\lambda(t) &> \lambda(0) e^{-t} \\
\frac{||\nabla^{-1}\theta||_{L^2}}{||\theta||_{L^2}} &\geq \frac{||\nabla^{-1}\theta_0||_{L^2}}{||\theta_0||_{L^2}} e^{-t} \\
||\nabla^{-1}\theta||_{L^2} &\geq \frac{||\nabla^{-1}\theta_0||_{L^2}}{||\theta_0||_{L^2}} e^{-t} ||\theta||_{L^2}
\end{aligned} \tag{3.2.10}$$

Using (3.2.9) in (3.2.10)

$$||\nabla^{-1}\theta||_{L^2} \geq ||\nabla^{-1}\theta_0||_{L^2} \exp \left[-t - \frac{1}{2Pe} \frac{||\nabla\theta_0||_{L^2}^2}{||\theta_0||_{L^2}^2} (e^{2t} - 1) \right]$$

3.2.2 Energy constrained $\|u\|_{L^\infty} = 1$

$$\begin{aligned} \|\nabla\theta\|_{L^2}^2 &= -2 \int_D \theta \Delta\theta \, d^d\mathbf{x} \\ &= Pe \int_D \theta (\partial_t\theta - \frac{1}{Pe} \Delta\theta) \, d^d\mathbf{x} - Pe \int_D \theta (\partial_t\theta + \frac{1}{Pe} \nabla\theta) \, d^d\mathbf{x} \end{aligned}$$

$$\begin{aligned} \frac{d}{dt} \|\theta\|_{L^2}^2 &= 2 \int_D \theta \partial_t\theta \, d^d\mathbf{x} \\ &= \int_D \theta (\partial_t\theta - \frac{1}{Pe} \Delta\theta) \, d^d\mathbf{x} + \int_D \theta (\partial_t\theta + \frac{1}{Pe} \Delta\theta) \, d^d\mathbf{x} \end{aligned}$$

and

$$\begin{aligned} \frac{d}{dt} \|\nabla\theta\|_{L^2}^2 &= -2 \int_D \partial_t\theta \Delta\theta \, d^d\mathbf{x} \\ &= Pe \int_D (\partial_t\theta - \frac{1}{Pe} \Delta\theta)^2 \, d^d\mathbf{x} - Pe \int_D (\partial_t\theta + \frac{1}{Pe} \Delta\theta)^2 \, d^d\mathbf{x} \end{aligned}$$

where we have used $-2ab = \frac{(a-b)^2 - (a+b)^2}{2}$.

Now consider

$$\begin{aligned} \frac{d}{dt} \left(\frac{\|\nabla\theta\|_{L^2}^2}{\|\theta\|_{L^2}^2} \right) &= \frac{1}{\|\theta\|_{L^2}^4} \left(\|\theta\|_{L^2}^2 \frac{d}{dt} \|\nabla\theta\|_{L^2}^2 - \frac{d}{dt} \|\theta\|_{L^2}^2 \|\nabla\theta\|_{L^2}^2 \right) \\ &= \frac{1}{\|\theta\|_{L^2}^2} \left[Pe \int_D (\partial_t\theta - \frac{1}{Pe} \Delta\theta)^2 \, d^d\mathbf{x} - Pe \int_D (\partial_t\theta + \frac{1}{Pe} \Delta\theta)^2 \, d^d\mathbf{x} \right] \\ &\quad - \frac{1}{\|\theta\|_{L^2}^4} \left[Pe \left(\int_D \theta (\partial_t\theta - \frac{1}{Pe} \Delta\theta) \, d^d\mathbf{x} \right)^2 - Pe \left(\int_D \theta (\partial_t\theta + \frac{1}{Pe} \Delta\theta) \, d^d\mathbf{x} \right)^2 \right] \end{aligned}$$

Using Hölder's inequality and $\partial_t + u \cdot \nabla\theta = k\Delta\theta$, we have

$$\frac{d}{dt} \left(\frac{\|\nabla\theta\|_{L^2}^2}{\|\theta\|_{L^2}^2} \right) \leq \frac{Pe}{\|\theta\|_{L^2}^2} \left[\int_D (u \cdot \nabla\theta)^2 \, d^d\mathbf{x} \right]$$

Using Holder's inequality:

$$\int_D (u \cdot \nabla\theta)^2 \, d^d\mathbf{x} \leq \|u\|_{L^\infty}^2 \int_D (\nabla\theta)^2 \, d^d\mathbf{x} = \|\nabla\theta\|_{L^2}^2$$

we have

$$\frac{d}{dt} \left(\frac{\|\nabla\theta\|_{L^2}^2}{\|\theta\|_{L^2}^2} \right) \leq Pe \frac{\|\nabla\theta\|_{L^2}^2}{\|\theta\|_{L^2}^2} \quad (3.2.11)$$

Using Gronwall's inequality, we have:

$$\begin{aligned} \left(\frac{\|\nabla\theta\|_{L^2}^2}{\|\theta\|_{L^2}^2} \right) &\leq \frac{\|\nabla\theta_0\|_{L^2}^2}{\|\theta_0\|_{L^2}^2} \exp(\text{Pe} \times t) \\ \left(\frac{\|\nabla\theta\|_{L^2}}{\|\theta\|_{L^2}} \right) &\leq \frac{\|\nabla\theta_0\|_{L^2}}{\|\theta_0\|_{L^2}} \exp\left(\frac{\text{Pe}}{2}t\right) \end{aligned} \quad (3.2.12)$$

Using (3.2.4), we have:

$$\frac{d}{dt} \|\theta\|_{L^2}^2 \geq \frac{-2}{\text{Pe}} \|\theta\|_{L^2}^2 \frac{\|\nabla\theta_0\|_{L^2}^2}{\|\theta_0\|_{L^2}^2} \exp(\text{Pe} \times t)$$

Using Gronwall's inequality, we have:

$$\|\theta\|_{L^2} \geq \|\theta_0\|_{L^2} \exp \left[-\frac{1}{\text{Pe}^2} \frac{\|\nabla\theta_0\|_{L^2}^2}{\|\theta_0\|_{L^2}^2} (e^{(\text{Pe} \times t)} - 1) \right] \quad (3.2.13)$$

From (3.2.2), then

$$\lambda(t) \geq \frac{\|\nabla^{-1}\theta\|_{L^2}}{\|\theta\|_{L^2}} \quad (3.2.14)$$

From $\|\nabla\theta\|_{L^2} \|\nabla^{-1}\theta\|_{L^2} \geq \|\theta\|_{L^2}^2$, we have

$$\frac{\|\nabla^{-1}\theta\|_{L^2}}{\|\theta\|_{L^2}} \geq \frac{\|\theta\|_{L^2}}{\|\nabla\theta\|_{L^2}}$$

Using (3.2.14), implying on the other hand

$$\lambda(t) \geq \frac{\|\nabla^{-1}\theta\|_{L^2}}{\|\theta\|_{L^2}} \geq \frac{\|\theta\|_{L^2}}{\|\nabla\theta\|_{L^2}} \geq \frac{\|\theta_0\|_{L^2}}{\|\nabla\theta_0\|_{L^2}} e^{(-\frac{\text{Pe}t}{2})} \quad (3.2.15)$$

and, on the other hand, using (3.2.13) in (3.2.15), we have:

$$\begin{aligned} \|\nabla^{-1}\theta\|_{L^2} &\geq \frac{\|\theta_0\|_{L^2}}{\|\nabla\theta_0\|_{L^2}} e^{(-\frac{\text{Pe}t}{2})} \|\theta\|_{L^2} \\ &\geq \frac{\|\theta_0\|_{L^2}^2}{\|\nabla\theta_0\|_{L^2}} \exp \left[\frac{-\text{Pe}t}{2} - \frac{1}{\text{Pe}^2} \frac{\|\nabla\theta_0\|_{L^2}^2}{\|\theta_0\|_{L^2}^2} (e^{\text{Pe}t} - 1) \right] \end{aligned}$$

3.2.3 Results

The analytical results of [LLN⁺12] show that, in the presence of diffusion, finite-time perfect mixing is impossible for L^∞ energy- and enstrophy-constrained flows.

Chapter 4

Anomalous Dissipation

The material in this chapter is drawn from the paper of Drivas, Elgindi, Iyer, and Jeong [DEIJ22], which establishes anomalous dissipation for passive scalar transport and non-uniqueness of weak solutions of the transport equation. The theorems, proofs, constructions, and discussion below reproduce results from that work. We refer the reader to [DEIJ22] for the original presentation.

This chapter examines anomalous dissipation in passive scalar transport, presenting the results of [DEIJ22]. The central equation is the advection-diffusion equation

$$\partial_t \theta^\kappa + u \cdot \nabla \theta^\kappa = k \Delta \theta^\kappa \quad (4.0.1)$$

posed on the d -dimensional torus $\mathbb{T}^d$, where θ^κ is a passive scalar, $k > 0$ is the molecular diffusivity, and u is a divergence-free advecting velocity field.

Testing (4.0.1) against θ^κ and integrating by parts yields

$$\begin{aligned} \theta^\kappa \partial_t \theta^\kappa + \theta^\kappa u \cdot \nabla \theta^\kappa &= k \theta^\kappa \Delta \theta^\kappa \\ \frac{1}{2} \int \frac{\partial |\theta^\kappa(t)|^2}{\partial t} &= -k \int_0^t |\nabla \theta^\kappa(s)|_{L^2}^2 \, ds \\ \frac{1}{2} |\theta^\kappa(t)|_{L^2}^2 &= \frac{1}{2} |\theta^\kappa(0)|_{L^2}^2 - k \int_0^t |\nabla \theta^\kappa(s)|_{L^2}^2 \, ds \end{aligned} \quad (4.0.2)$$

The identity (4.0.2) shows that the integrated quantity $k \int_0^t |\nabla \theta^\kappa(s)|_{L^2}^2 \, ds$ precisely measures the L^2 energy lost to diffusion up to time t . The stirring action of the velocity field sharpens scalar gradients, which are then dissipated by diffusion. A particularly striking phenomenon arises when this dissipation remains strictly positive as $\kappa \rightarrow 0$: there exists a constant $\chi > 0$, independent of κ , such that

$$k \int_0^t |\nabla \theta^\kappa(s)|_{L^2}^2 \, ds \geq \chi > 0.$$

This persistence of energy loss in the inviscid limit is known as *anomalous dissipation*.

4.1 Notation

The proofs of the theorems in this chapter are presented in two spatial dimensions. Without loss of generality, let $T = 1$ and let the initial value θ_0 be mean zero:

$$\int_{\mathbb{T}^d} \theta_0(x) \, dx = 0.$$

$$|\theta|_{L^2} = \left(\int_{\mathbb{T}^2} |\theta|^2 dx \right)^{1/2}, \quad |\theta|_{\dot{H}^1}^2 = \sum_{i=1}^2 |\partial_i \theta|_{L^2}^2, \quad \text{and} \quad |\theta|_{\dot{H}^2}^2 = \sum_{i,j=1}^2 |\partial_i \partial_j \theta|_{L^2}^2,$$

for any function $\theta: \mathbb{T}^2 \rightarrow \mathbb{R}$. Also, define $|\theta|_{\dot{W}^{1,\infty}} = \max\{|\partial_1 \theta|_{L^\infty}, |\partial_2 \theta|_{L^\infty}\}$. With these conventions,

$$|\theta|_{\dot{H}^1}^2 \leq |\theta|_{L^2} |\theta|_{\dot{H}^2},$$

and

$$|\theta|_{H^2}^2 = |\theta|_{\dot{H}^2}^2 + |\theta|_{\dot{H}^1}^2 + |\theta|_{L^2}^2, \quad |\theta|_{H^1}^2 = |\theta|_{\dot{H}^1}^2 + |\theta|_{L^2}^2, \quad |\theta|_{W^{1,\infty}} = \max\{|\theta|_{L^\infty}, |\theta|_{\dot{W}^{1,\infty}}\}.$$

Throughout, the notation $A(t) \approx B(t)$ means that $A(t)/B(t)$ is bounded above and below by positive absolute constants, while $A \ll B$ means $\lim_{t \rightarrow 1} A/B = 0$.

4.2 Criterion for Anomalous Dissipation

This section presents three criteria from [DEIJ22] that guarantee anomalous dissipation. The proof of Theorem 3 involves comparing θ , the solution of the transport equation, to θ^κ , the solution of the advection-diffusion equation.

Proposition 4.2.1 ([DEIJ22, Proposition 1.1]). *Let $T > 0$, assume $u \in L_{\text{loc}}^\infty([0, T]; W^{1,\infty}(\mathbb{T}^d))$ is divergence free, and let θ^κ and θ be solutions to (4.0.1) and (4.3.4) respectively with the same, κ independent, mean zero initial data $\theta_0 \in H^2(\mathbb{T}^d)$.*

If

$$\lim_{t \rightarrow T} \int_0^t |\nabla \theta|_{L^2}^2 ds = +\infty, \quad \text{and} \quad |\theta(t)|_{\dot{H}^1}^2 \geq c(|\theta(t)|_{\dot{H}^2} + |\theta^\kappa(t)|_{\dot{H}^2}) |\theta_0|_{L^2} \quad (4.2.1)$$

for all $t \in [0, T)$, for a fixed constant $c \in (0, 1)$ independent of κ and t , then

$$\kappa \int_0^T |\nabla \theta^\kappa|_{L^2}^2 dt \geq \left(\frac{c}{2}\right)^4 |\theta_0|_{L^2}^2.$$

The next two results involve conditions on the inviscid equation alone.

Proposition 4.2.2 ([DEIJ22, Proposition 1.2]). *If*

$$\int_0^T |\nabla \theta(s)|_{L^2}^2 ds = +\infty \quad \text{and} \quad |\theta(t)|_{H^{-1}} \leq C \frac{|\theta_0|_{L^2}^2}{|\theta(t)|_{\dot{H}^1}}, \quad (4.2.2)$$

for all $t \in [0, T)$ and some constant $C \geq 1$ independent of κ and t , then

$$\kappa \int_0^T |\nabla \theta^\kappa|_{L^2}^2 dt \geq \frac{1}{64C^2} |\theta_0|_{L^2}^2.$$

Recall that $|\theta(t)|_{H^{-1}}$ measures the degree to which $\theta(t)$ is mixed. Proposition 4.2.2 thus creates a direct link between mixing and anomalous dissipation.

Proposition 4.2.3 below provides a condition for anomalous dissipation that requires only growth of positive norms of θ , a criterion weaker than mixing.

Proposition 4.2.3 ([DEIJ22, Proposition 1.3]). *If*

$$|\theta(t)|_{\dot{H}^2} \leq \frac{C|\theta(t)|_{\dot{H}^1}^2}{|\theta_0|_{L^2}}, \quad \frac{1}{C(T-t)} \leq \frac{|\theta(t)|_{\dot{H}^1}}{|\theta_0|_{L^2}}, \quad \text{and} \quad |\nabla u(t)|_{L^\infty} \leq \frac{C}{(T-t)}, \quad (4.2.3)$$

for all $t \in [0, T)$, and some constant $C \geq 1$, independent of κ and t , then there exists a $\chi > 0$ depending only on C

$$\kappa \int_0^T |\nabla \theta^\kappa|_{L^2}^2 dt > \chi |\theta_0|_{L^2}^2.$$

4.3 Results

The central theorems of Drivas, Elgindi, Iyer, and Jeong [DEIJ22] are as follows.

Theorem 3 (Universal rate near Harmonics [DEIJ22, Theorem 1.1]). *Fix $T > 0$, $d \geq 2$, and $\alpha \in [0, 1)$, and let*

$$\Psi \stackrel{\text{def}}{=} \{\sin(Mx) \sin(Ly), \sin(Mx) \cos(Ly), \sin(Ly) \cos(Mx), \cos(Mx) \cos(Ly)\}_{M \geq 0, L \geq 0} - \{0, 1\}.$$

There exists absolute constants $\varepsilon_\alpha, \chi_\alpha > 0$, and a divergence-free velocity field

$$u \in C^\infty([0, T] \times \mathbb{T}^d) \cap L^1([0, T]; C^\alpha(\mathbb{T}^d)) \cap L^\infty([0, T] \times \mathbb{T}^d), \quad (4.3.1)$$

such that the following holds: If $\theta_0 \in H^2(\mathbb{T}^d)$ is mean zero, and there exists $\lambda > 0$ and $\psi \in \Psi$ such that

$$|\theta_0 - \lambda \psi|_{L^2} \leq \varepsilon_\alpha |\theta_0|_{L^2},$$

then

$$\kappa \int_0^T |\nabla \theta^\kappa|_{L^2}^2 dt \geq \chi_\alpha |\theta_0|_{L^2}^2. \quad (4.3.2)$$

For H^2 initial data, a slight modification of the velocity field above also produces anomalous dissipation, but the velocity field and dissipation rate then depend on the data.

Theorem 4 (Data dependent rate and velocity field [DEIJ22, Theorem 1.2]). *Fix $T > 0$, $d \geq 2$, $\alpha \in [0, 1)$, and a mean-zero $\theta_0 \in H^2(\mathbb{T}^d)$. There exists a divergence-free velocity field*

$$u \in C^\infty([0, T] \times \mathbb{T}^d) \cap L^1([0, T]; C^\alpha(\mathbb{T}^d)) \cap L^\infty([0, T] \times \mathbb{T}^d),$$

and $\chi_\alpha(\theta_0) > 0$ so that we have

$$\kappa \int_0^T |\nabla \theta^\kappa|_{L^2}^2 dt \geq \chi_\alpha(\theta_0) |\theta_0|_{L^2}^2. \quad (4.3.3)$$

Theorem 5 (Non-uniqueness of the transport equation [DEIJ22, Theorem 1.3]). *Fix $T > 0$, $d \geq 2$, $\alpha \in [0, 1)$ and a mean-zero $\theta_0 \in H^2$. Let u_p be the divergence-free velocity field from Theorem 4. Let u , defined on $[0, 2T]$, be*

$$u(t) = \begin{cases} u_p(t) & t \in [0, T], \\ -u_p(2T - t) & t \in [T, 2T]. \end{cases}$$

Then there are at least two weak solutions $\theta, \bar{\theta} \in C_w([0, 2T]; L^2(\mathbb{T}^d))$ of the transport equation

$$\partial_t \theta + u \cdot \nabla \theta = 0 \quad (4.3.4)$$

with initial data θ_0 .

Theorem 5 is proved by constructing one solution using time reversibility and another as a vanishing viscosity limit; this construction is presented in Section 4.6.

4.4 Criteria for Anomalous Dissipation

This section reproduces the proofs of Propositions 4.2.1–4.2.3 from [DEIJ22]. The first result will also be used in the proof of Theorem 3.

4.4.1 Non-viscous Growth Criterion with an Assumption on $|\theta^\kappa|_{H^2}$

Proof of Proposition 4.2.1. Normalise so that $|\theta_0|_{L^2} = 1$. The key starting point is the energy identity for the difference $\theta - \theta^\kappa$:

$$\frac{1}{2} \partial_t |\theta - \theta^\kappa|_{L^2}^2 = \kappa \int \Delta \theta^\kappa (\theta^\kappa - \theta) dx = -\kappa |\theta^\kappa|_{\dot{H}^1}^2 + \kappa \int \nabla \theta^\kappa \cdot \nabla \theta dx.$$

Integrating in time and applying Cauchy-Schwarz gives

$$\frac{1}{2} |\theta - \theta^\kappa|_{L^2}^2(t) \leq \left(\kappa \int_0^1 |\theta^\kappa|_{\dot{H}^1}^2 ds \right)^{1/2} \left(\kappa \int_0^t |\theta|_{\dot{H}^1}^2 ds \right)^{1/2}, \quad (4.4.1)$$

for all $t < 1$.

Suppose for contradiction that there is a sequence $\kappa_k \rightarrow 0$ with

$$\delta_k \stackrel{\text{def}}{=} \kappa_k \int_0^1 |\theta^{\kappa_k}|_{\dot{H}^1}^2 dt < \chi \quad \text{as} \quad k \rightarrow \infty, \quad (4.4.2)$$

where $\chi \stackrel{\text{def}}{=} (c/2)^4$. Define $T_k < 1$ by

$$\kappa_k \int_0^{T_k} |\theta|_{\dot{H}^1}^2 dt = 1. \quad (4.4.3)$$

Since $|\theta|_{\dot{H}^1}^2 \rightarrow +\infty$ by hypothesis, $T_k \rightarrow 1$ as $k \rightarrow \infty$. Substituting (4.4.2) and (4.4.3) into (4.4.1) yields

$$\sup_{t \leq T_k} |\theta - \theta^{\kappa_k}|_{L^2} \leq \sqrt{2} \delta_k^{1/4} \leq \sqrt{2} \chi^{1/4}.$$

Using the interpolation inequality together with assumption (4.2.1) for $t \leq T_k$,

$$|\theta - \theta^{\kappa}|_{\dot{H}^1}^2 \leq |\theta - \theta^{\kappa}|_{L^2} |\theta - \theta^{\kappa}|_{\dot{H}^2} \leq \frac{\sqrt{2} \chi^{1/4}}{c} |\theta|_{\dot{H}^1}^2,$$

so that $|\theta - \theta^{\kappa}|_{\dot{H}^1} \leq \frac{2^{1/4} \chi^{1/8}}{\sqrt{c}} |\theta|_{\dot{H}^1}$. The reverse triangle inequality then gives

$$|\theta^{\kappa}|_{\dot{H}^1} \geq \left| |\theta|_{\dot{H}^1} - |\theta^{\kappa} - \theta|_{\dot{H}^1} \right| \geq \left(1 - \frac{2^{1/4} \chi^{1/8}}{\sqrt{c}} \right) |\theta|_{\dot{H}^1}. \quad (4.4.4)$$

Integrating and applying (4.4.3):

$$\kappa_k \int_0^{T_k} |\theta^{\kappa}|_{\dot{H}^1}^2 dt \geq \kappa_k \left(1 - \frac{2^{1/4} \chi^{1/8}}{\sqrt{c}} \right)^2 \int_0^{T_k} |\theta|_{\dot{H}^1}^2 dt = \left(1 - \frac{2^{1/4} \chi^{1/8}}{\sqrt{c}} \right)^2. \quad (4.4.5)$$

This contradicts (4.4.2) provided χ satisfies

$$\chi \leq \left(1 - \frac{2^{1/4} \chi^{1/8}}{\sqrt{c}} \right)^2. \quad (4.4.6)$$

With $\chi = (c/2)^4$, inequality (4.4.6) reduces to

$$\left(\frac{c}{4} \right)^2 \leq 1 - \left(\frac{1}{4} \right)^{1/4},$$

which holds since $c < 1$ and $1/16 < 1 - (1/2)^{1/4}$. □

4.4.2 Inviscid Mixing Criterion

Proof of Proposition 4.2.2. Normalise so that $|\theta_0|_{L^2} = 1$. The argument is a refinement of the proof of Proposition 4.2.1: assume for contradiction that (4.4.2) holds with $\chi \stackrel{\text{def}}{=} \frac{1}{2^6 C^2}$, and define T_k as in (4.4.3). The bound (4.4.1) then gives

$$\sup_{t \leq T_k} |\theta - \theta^{\kappa_k}|_{L^2} \leq \sqrt{2} \chi^{1/4} = \frac{1}{2} C^{-1/2} \leq \frac{1}{2},$$

where the last inequality uses $C \geq 1$ as required by (4.2.2).

For any $N \in \mathbb{N}$, let $\mathbf{P}_{>N}$ denote the projection onto Fourier modes with frequency greater than N . The triangle inequality together with $|\theta|_{L^2} = 1$ yields

$$\begin{aligned} |\mathbf{P}_{>N} \theta^\kappa|_{L^2} &\geq |\mathbf{P}_{>N} \theta|_{L^2} - |\mathbf{P}_{>N} (\theta - \theta^\kappa)|_{L^2} \geq |\mathbf{P}_{>N} \theta|_{L^2} - \frac{1}{2} \\ &\geq 1 - N^2 |\theta|_{H^{-1}}^2 - \frac{1}{2} \geq \frac{1}{2} - \frac{C^2 N^2}{|\theta(t)|_{H^1}^2}, \end{aligned}$$

where C is the constant from (4.2.2). Setting $\lambda(t) \stackrel{\text{def}}{=} |\theta(t)|_{H^1}^2$ and choosing $N^2 = \lambda(t)/(2C)^2$ makes the right side equal to $1/4$, so

$$|\mathbf{P}_{>N} \theta^\kappa|_{L^2} \geq \frac{1}{4} \quad \implies \quad |\nabla \theta^\kappa|_{L^2}^2 \geq \frac{\lambda(t)}{16C^2}.$$

Integrating,

$$\kappa \int_0^{T_k} |\nabla \theta^\kappa|_{L^2}^2 dt \geq \frac{1}{16C^2} = 4\chi,$$

which contradicts (4.4.2) and completes the proof. $\square$

4.5 Construction of the Example

In this section, following [DEIJ22], Theorems 3 and 4 are established by exhibiting an explicit velocity field satisfying the conditions of Proposition 4.2.1. For simplicity, $T = 1$ and $d = 2$ are assumed throughout this section. While the theorems state regularity (4.3.1), the constructed velocity belongs to the slightly stronger class

$$u \in L_{\text{loc}}^\infty([0, 1]; W^{2,\infty}(\mathbb{T}^2)) \cap L^1([0, 1]; C^\alpha(\mathbb{T}^2)) \cap L^\infty([0, 1] \times \mathbb{T}^2);$$

the improvement to C^∞ is described in the subsequent remark.

The key building block of the construction is a smoothed-out sawtooth function. Given $\frac{\pi}{2} > \varepsilon > 0$, we define $S_\varepsilon : \mathbb{T} = (\mathbb{R}/2\pi\mathbb{Z}) \rightarrow \mathbb{R}$ to be odd with respect to 0, even with respect to $\frac{\pi}{2}$, and

$$S_\varepsilon(x) = \begin{cases} x & 0 \leq x \leq \frac{\pi}{2} - \varepsilon \\ x - \frac{1}{2\varepsilon}(x - \frac{\pi}{2} + \varepsilon)^2 & \frac{\pi}{2} - \varepsilon < x \leq \frac{\pi}{2} \end{cases}$$

One verifies directly that $S_\varepsilon \in W^{2,\infty}(\mathbb{T})$ with

$$|S'_\varepsilon|_{L^\infty} \leq 1 \quad \text{and} \quad |S''_\varepsilon|_{L^\infty} \leq \frac{1}{\varepsilon}.$$

The velocity field is built from three sequences: time steps $\{t_j\}_{j \in \mathbb{N}}$ that are decreasing and summable, regularization parameters ε_j that are also decreasing and summable, and frequencies $\{N_j\}_{j \in \mathbb{N}}$ that grow rapidly. From these, one defines measure-preserving shear maps

$$\mathcal{T}_j(x, y) = \begin{cases} (x + t_j S_{\varepsilon_j}(N_j y), y) & j \text{ odd} \\ (x, y + t_j S_{\varepsilon_j}(N_j x)) & j \text{ even,} \end{cases}$$

and their compositions $\mathcal{U}_j \stackrel{\text{def}}{=} \mathcal{T}_1 \circ \mathcal{T}_2 \circ \mathcal{T}_3 \circ \cdots \circ \mathcal{T}_j$. Setting $T_j = \sum_{i=0}^j t_i$ with $t_0 = 0$, one can verify that $\theta_0 \circ \mathcal{U}_j = \theta(T_j)$, where $\theta(t)$ solves the inviscid equation

$$\partial_t \theta + u \cdot \nabla \theta = 0, \quad \theta(t = 0) = \theta_0,$$

with the alternating-shear velocity field, for $t \in [T_i, T_{i+1})$,

$$u(t, x, y) = \begin{cases} \begin{pmatrix} S_{\varepsilon_i}(N_i y) \\ 0 \end{pmatrix} & i \text{ even,} \\ \begin{pmatrix} 0 \\ S_{\varepsilon_i}(N_i x) \end{pmatrix} & i \text{ odd.} \end{cases} \quad (4.5.1)$$

The remaining sections verify the conditions of Proposition 4.2.1 for this velocity field. For small $\alpha > 0$, a minor technical modification sets $u(t)$ to zero for $t \leq T_j$ with $j < 1/\alpha$; see Lemma 4.5.2 for details.

Figure 4.1: Fix $\alpha = 1$ and ε_i and N_i as above. The two left panels depict shear profiles, and two right panels represent contour plots of the corresponding stream function with velocity vectors superimposed. The two top panels correspond to $t_2 = 2^{-2}$, and the two Bottom panels correspond to $t_3 = 2^{-3}$.

Remark 4.5.1. The velocity constructed above belongs to $L^\infty_{\text{loc}}([0, T]; W^{1,\infty}(\mathbb{T}^2))$ rather than $C^\infty([0, 1] \times \mathbb{T}^2)$ as stated in Theorem 3. To achieve full smoothness, one replaces the C^2 profile S_ε by a C^∞ one and multiplies each shear by an amplitude function that ramps smoothly on and off within each time interval. These ramp windows can be made arbitrarily short, shrinking as j increases, so the key quantitative estimates remain intact.

4.5.1 Inviscid Bounds

The following lemma is established in [DEIJ22].

Lemma 4.5.2. *For any $\alpha > 0$, let*

$$t_j = 2^{-j}, \quad N_j = 2^{(1+\alpha)j}, \quad \varepsilon_j = \exp\left(-30\left(1 + \frac{1}{2^\alpha - 1}\right)\right) \cdot 2^{-2j}.$$

Assume that θ_0 is given by one of the following trigonometric functions:

$$\sin(Mx)\sin(Ly), \quad \sin(Mx)\cos(Ly), \quad \cos(Lx)\sin(My), \quad \cos(Lx)\cos(My)$$

for some integers $M \geq 1$ and $L \geq 0$. Define $\theta_j(x, y) = \theta_0 \circ \mathcal{U}_j$ where

$$\mathcal{U}_j \stackrel{\text{def}}{=} \text{Id}, \quad j \leq \frac{1}{\alpha}$$

and

$$\mathcal{U}_j \stackrel{\text{def}}{=} \mathcal{T}_{J(\alpha)} \circ \cdots \circ \mathcal{T}_j, \quad j \geq J(\alpha) \stackrel{\text{def}}{=} \left\lfloor \frac{1}{\alpha} \right\rfloor + 1.$$

Here $\lfloor \frac{1}{\alpha} \rfloor$ denotes the largest integer not exceeding $\frac{1}{\alpha}$. Then, θ_j satisfy:

$$|\theta_j|_{H^1} \geq c_\alpha 2^{\frac{\alpha j(j+1)}{2}} |\theta_0|_{H^1}, \quad |\theta_j|_{W^{1,\infty}} \leq C_\alpha |\theta_j|_{H^1}, \quad |\theta_j|_{H^2} \leq C_\alpha |\theta_j|_{H^1}^2,$$

for some constants $C_\alpha, c_\alpha > 0$ independent of j .

Remark 4.5.3. It is easy to see that the choice of t_j, N_j, ε_j in the above gives us $|\nabla u(t)|_{L^\infty} \approx \frac{1}{(1-t)^2}$ while $|\theta(t)|_{H^1}^2 \approx |\theta(t)|_{H^2} \gg \frac{1}{(1-t)^2}$ where u is defined as in (4.5.1).

Proof. Take $\theta_0 = \sin(Mx)\sin(Ly)$ with $M \geq L$ for concreteness; the argument for the other listed trigonometric functions is virtually identical, and the case $L > M$ is addressed at the end. A direct computation shows that

$$|\theta_0|_{\dot{W}^{1,\infty}} \leq 4|\theta_0|_{\dot{H}^1}, \quad |\theta_0|_{\dot{H}^2} \leq 4|\theta_0|_{\dot{H}^1}^2.$$

Fix $j \in \mathbb{N} \cup \{0\}$ and set $i_j = j - 1 \bmod 2$ and $i_{j+1} = j \bmod 2$. For j odd, differentiating $\theta_{j+1}(x, y) = \theta_j(x + t_j S_{\varepsilon_j}(N_j y), y)$ gives:

$$\begin{aligned} \partial_2 \theta_{j+1}(x, y) &= \partial_2 \theta_j(x + t_j S_{\varepsilon_j}(N_j y), y) + t_j N_j S'_{\varepsilon_j}(N_j y) \partial_1 \theta_j(x + t_j S_{\varepsilon_j}(N_j y), y), \\ \partial_1 \theta_{j+1}(x, y) &= \partial_1 \theta_j(x + t_j S_{\varepsilon_j}(N_j y), y). \end{aligned}$$

Moreover,

$$\begin{aligned}\partial_{11}\theta_{j+1}(x, y) &= \partial_{11}\theta_j(x + t_j S_{\varepsilon_j}(N_j y), y), \\ \partial_{12}\theta_{j+1}(x, y) &= \partial_{12}\theta_j(x + t_j S_{\varepsilon_j}(N_j y), y) + t_j N_j S'_{\varepsilon_j}(N_j y) \partial_{11}\theta_j(x + t_j S_{\varepsilon_j}(N_j y), y), \\ \partial_{22}\theta_{j+1}(x, y) &= \partial_{22}\theta_j(x + t_j S_{\varepsilon_j}(N_j y), y) + t_j N_j^2 S''_{\varepsilon_j}(N_j y) \partial_{11}\theta_j(x + t_j S_{\varepsilon_j}(N_j y), y) \\ &\quad + (t_j N_j)^2 (S'_{\varepsilon_j}(N_j y))^2 \partial_{11}\theta_j(x + t_j S_{\varepsilon_j}(N_j y), y).\end{aligned}$$

When j is even, analogous formulas hold with the roles of x and y interchanged. These derivative identities are the foundation for both the upper and lower bounds on $|\theta_{j+1}|$.

Upper Bounds: Since $|S'_{\varepsilon}|_{L^\infty} \leq 1$, the Lipschitz bound on S_ε immediately gives

$$\begin{aligned}|\theta_{j+1}|_{\dot{W}^{1,\infty}} &\leq t_j N_j |\theta_j|_{\dot{W}^{1,\infty}} + |\theta_j|_{\dot{W}^{1,\infty}} \\ &\leq t_j N_j |\theta_j|_{\dot{W}^{1,\infty}} \left(1 + \frac{1}{t_j N_j}\right).\end{aligned}$$

The same reasoning gives the analogous bound for the $\dot{H}^1$ norm:

$$|\theta_{j+1}|_{\dot{H}^1} \leq t_j N_j |\theta_j|_{\dot{H}^1} \left(1 + \frac{1}{t_j N_j}\right).$$

With $t_j = 2^{-j}$ and $N_j = 2^{(1+\alpha)j}$, the accumulated product over all steps satisfies

$$\prod_{j=1}^{\infty} \left(1 + \frac{1}{t_j N_j}\right) = \prod_{j=1}^{\infty} \left(1 + 2^{-\alpha j}\right) \leq \exp\left(\frac{1}{2^\alpha - 1}\right),$$

by the bound $\log(1+x) \leq x$. Telescoping the one-step estimates therefore yields

$$|\theta_{j+1}|_{\dot{W}^{1,\infty}} \leq \exp\left(\frac{1}{2^\alpha - 1}\right) \cdot 2^{\frac{\alpha j(j+1)}{2}} |\theta_0|_{\dot{W}^{1,\infty}}, \quad |\theta_{j+1}|_{\dot{H}^1} \leq \exp\left(\frac{1}{2^\alpha - 1}\right) \cdot 2^{\frac{\alpha j(j+1)}{2}} |\theta_0|_{\dot{H}^1}.$$

Lower Bounds: The derivative $|S'_{\varepsilon_j}(z)|$ equals 1 everywhere except in the smoothing region $|z - \frac{\pi}{2}| < \varepsilon_j$. The error from this region is controlled by the $\dot{W}^{1,\infty}$ norm:

$$\begin{aligned}|\partial_{i_{j+1}}\theta_{j+1}|_{L^2} &\geq t_j N_j |\partial_{i_j}\theta_j|_{L^2} - \sqrt{\varepsilon_j} t_j N_j |\theta_j|_{\dot{W}^{1,\infty}} - |\partial_{i_{j+1}}\theta_j|_{L^2} \\ &= t_j N_j |\partial_{i_j}\theta_j|_{L^2} \left(1 - \sqrt{\varepsilon_j} \frac{|\theta_j|_{\dot{W}^{1,\infty}}}{|\partial_{i_j}\theta_j|_{L^2}} - \frac{1}{t_j N_j} \frac{|\partial_{i_{j-1}}\theta_j|_{L^2}}{|\partial_{i_j}\theta_j|_{L^2}}\right).\end{aligned}$$

Since the shear only acts in the x -direction when j is odd, the i_j -derivative is unchanged: $|\partial_{i_j}\theta_{j+1}|_{L^2} = |\partial_{i_j}\theta_j|_{L^2}$. Combining both estimates,

$$|\partial_{i_{j+1}}\theta_{j+1}|_{L^2} \geq t_j N_j |\partial_{i_j}\theta_j|_{L^2} \left(1 - \sqrt{\varepsilon_j} \frac{|\theta_j|_{\dot{W}^{1,\infty}}}{|\partial_{i_j}\theta_j|_{L^2}} - \frac{1}{t_j N_j} \frac{|\partial_{i_{j-1}}\theta_{j-1}|_{L^2}}{|\partial_{i_j}\theta_j|_{L^2}}\right),$$

where $\theta_{-1} \equiv 0$. The ratios

$$A_j \stackrel{\text{def}}{=} \frac{|\partial_{i_j} \theta_j|_{L^2}}{|\partial_{i_{j+1}} \theta_{j+1}|_{L^2}}, \quad R_{j+1} \stackrel{\text{def}}{=} \frac{|\theta_{j+1}|_{\dot{W}^{1,\infty}}}{|\partial_{i_{j+1}} \theta_{j+1}|_{L^2}}$$

then satisfy the recursive inequalities

$$\begin{aligned} A_j &\leq 2^{-\alpha j} \left(1 - \sqrt{\varepsilon_j} R_j - 2^{-\alpha j} A_{j-1}\right)^{-1}, \\ R_{j+1} &\leq R_j \left(1 + 2^{-\alpha j}\right) \left(1 - \sqrt{\varepsilon_j} R_j - 2^{-\alpha j} A_{j-1}\right)^{-1}. \end{aligned}$$

With $\varepsilon_j = \exp(-30(1 + \frac{1}{2^\alpha - 1}))2^{-2j}$, a bootstrap argument shows:

$$A_j \leq 2 \cdot 2^{-\alpha j}, \quad R_j \leq \exp(10(1 + \frac{1}{2^\alpha - 1})).$$

Under these bounds, the recursion for A_j simplifies to

$$A_j \leq 2^{-\alpha j} \left(\frac{9}{10} - 2^{-\alpha j} A_{j-1}\right)^{-1}. \quad (4.5.2)$$

To initialise the bootstrap, take $\alpha = 1$ and compute explicitly with $\theta_0 = \sin(Mx) \sin(Ly)$:

$$\theta_1(x, y) = \sin(M(x + t_1 S_{\varepsilon_1}(N_1 y))) \cos(Ly),$$

so that

$$\partial_y \theta_1 = t_1 N_1 S'_{\varepsilon_1}(N_1 y) M \cos(M(x + t_1 S_{\varepsilon_1}(N_1 y))) \cos(Ly) - L \sin(M(x + t_1 S_{\varepsilon_1}(N_1 y))) \sin(Ly).$$

The two terms are L^2 -orthogonal, which gives

$$A_0 \leq 2^{-1} (1 - \sqrt{\varepsilon_0} R_0)^{-1} < \frac{3}{5},$$

since $R_0 = |\partial_x \theta_0|_{L^\infty} / |\partial_x \theta_0|_{L^2}$ is uniformly bounded for trigonometric initial data. For general $\alpha > 0$, one obtains the same bound $A_0 = A_{J(\alpha)-1} < \frac{3}{5}$ by an analogous calculation; the remaining trigonometric cases require only minor modifications. Substituting $A_{J(\alpha)-1} < \frac{3}{5}$ into (4.5.2) gives $A_{J(\alpha)} < 1$, and from this point the bootstrap closes: $A_j < 2 \cdot 2^{-\alpha j}$ for all $j \geq J(\alpha) + 1$.

The bound $R_j < \exp(10(1 + \frac{1}{2^\alpha - 1}))$ is maintained by the recursion

$$R_{j+1} \leq R_j (1 + 2^{-\alpha j}) (1 - \frac{1}{100} 2^{-j} - 2^{-\alpha j} A_{j-1})^{-1},$$

together with $\frac{1}{1-x} < 1 + 3x$ for small x . Using the summability of the correction factors:

$$\prod_{j=J(\alpha)}^{\infty} (1 + 2^{-\alpha j}) < \exp\left(\frac{1}{2^\alpha - 1}\right),$$

$$\prod_{j=J(\alpha)}^{\infty} (1 - \frac{1}{100}2^{-j} - 2^{-\alpha j}A_{j-1})^{-1} \leq \prod_{j=J(\alpha)}^{\infty} \left(1 + \frac{3}{100}2^{-j} + 3 \cdot 2^{-\alpha j}A_{j-1}\right),$$

where the last product is bounded by $\exp(10(1 + \frac{1}{2^{\alpha-1}}))$ after substituting the bootstrap bound on A_j . This closes the bootstrap and gives

$$|\partial_{ij}\theta_j|_{L^2} \geq c_\alpha 2^{\frac{\alpha j(j+1)}{2}} |\partial_x \theta_0|_{L^2}.$$

$\dot{H}^2$ Bound: The second-order derivative estimates yield

$$\begin{aligned} |\theta_{j+1}|_{\dot{H}^2} &\leq t_j^2 N_j^2 |\theta_j|_{\dot{H}^2} (1 + 2(t_j N_j)^{-1} + (t_j N_j)^{-2}) + |\theta_j|_{\dot{H}^1} \frac{t_j}{\varepsilon_j} N_j^2, \\ &\leq 2^{2\alpha j} |\theta_j|_{\dot{H}^2} \left(1 + 2 \cdot 2^{-\alpha j} + 2^{-2\alpha j} + C_\alpha \frac{|\theta_j|_{\dot{H}^1}}{|\theta_j|_{\dot{H}^2}} 2^{3j}\right). \end{aligned}$$

Invoking the Sobolev interpolation $|\theta_j|_{\dot{H}^2} \geq |\theta_j|_{\dot{H}^1}^2$ and the $\dot{H}^1$ lower bound,

$$|\theta_{j+1}|_{\dot{H}^2} \leq 2^{2\alpha j} |\theta_j|_{\dot{H}^2} \left(1 + 2 \cdot 2^{-\alpha j} + 2^{-2\alpha j} + C_\alpha 2^{-\frac{\alpha j(j+1)}{2}} 2^{3j}\right).$$

The correction series satisfies

$$\sum_{j=0}^{\infty} (2 \cdot 2^{-\alpha j} + 2^{-2\alpha j} + C_\alpha 2^{-\frac{\alpha j(j+1)}{2}} 2^{3j}) = c_\alpha < \infty,$$

so that after telescoping

$$|\theta_{j+1}|_{\dot{H}^2} \leq 2^{\alpha j(j+1)} e^{c_\alpha} |\theta_0|_{\dot{H}^2}.$$

This closes the argument for $M \geq L$. When $L > M$, the analogous computation at $j = 1$ (with $\alpha = 1$) shows $|\theta_1|_{\dot{H}^1} \approx |\partial_y \theta_1|_{L^2}$ and

$$|\theta_1|_{\dot{W}^{1,\infty}} \leq 10 |\theta_1|_{\dot{H}^1}, \quad |\theta_0|_{\dot{H}^2} \leq 10 |\theta_0|_{\dot{H}^1}^2,$$

so the same bootstrap applies starting from $j = 1$ rather than $j = 0$. $\square$

4.5.2 Viscous Bounds

To complete the proof of anomalous dissipation, the following lemma from [DEIJ22] is also needed.

Lemma 4.5.4. *Assume that t_j, N_j, ε_j , and θ_0 are given as in Lemma 4.5.2. With $u(t)$ defined as in (4.5.1), the solution θ^κ to the system (4.0.1) with initial data θ_0 satisfies*

$$|\theta^\kappa(t)|_{H^2} \leq C |\theta(t)|_{H^1}^2$$

for some universal constant C independent of κ of t , where $\theta(t)$ is the inviscid solution.

Proof. Take $\theta_0 = \sin(Mx)\sin(Ly)$ with $M \geq L$; the other trigonometric cases are analogous. Recall that $T_j = \sum_{i=0}^j t_i$. Since the velocity alternates between horizontal and vertical shears, it suffices to prove the H^2 bound on each sub-interval $[T_j, T_{j+1}]$ separately. For j even, the relevant equation is

$$\partial_t \theta^\kappa + S_{\varepsilon_j}(N_j y) \partial_x \theta^\kappa = \kappa \Delta \theta^\kappa, \quad t \in [T_j, T_{j+1}]$$

(the case j odd is virtually identical with x and y exchanged). Differentiating in all multi-indices of order at most 2 gives:

$$\begin{aligned} \partial_t \theta^\kappa + S_{\varepsilon_j}(N_j y) \partial_x \theta^\kappa &= \kappa \Delta \theta^\kappa, \\ \partial_t \partial_x \theta^\kappa + S_{\varepsilon_j}(N_j y) \partial_{xx} \theta^\kappa &= \kappa \Delta \partial_x \theta^\kappa, \\ \partial_t \partial_{xx} \theta^\kappa + S_{\varepsilon_j}(N_j y) \partial_{xxx} \theta^\kappa &= \kappa \Delta \partial_{xx} \theta^\kappa, \\ \partial_t \partial_y \theta^\kappa + S_{\varepsilon_j}(N_j y) \partial_x \partial_y \theta^\kappa + N_j S'_{\varepsilon_j}(N_j y) \partial_x \theta^\kappa &= \kappa \Delta \partial_y \theta^\kappa, \\ \partial_t \partial_{xy} \theta^\kappa + S_{\varepsilon_j}(N_j y) \partial_x \partial_{xy} \theta^\kappa + N_j S'_{\varepsilon_j}(N_j y) \partial_{xx} \theta^\kappa &= \kappa \Delta \partial_{xy} \theta^\kappa, \\ \partial_t \partial_{yy} \theta^\kappa + S_{\varepsilon_j}(N_j y) \partial_x \partial_{yy} \theta^\kappa + 2N_j S'_{\varepsilon_j}(N_j y) \partial_{xy} \theta^\kappa + N_j^2 S''_{\varepsilon_j}(N_j y) \partial_x \theta^\kappa &= \kappa \Delta \partial_{yy} \theta^\kappa. \end{aligned}$$

The first three equations (for θ^κ , $\partial_x \theta^\kappa$, $\partial_{xx} \theta^\kappa$) have no commutator terms, so the L^2 dissipation is monotone and

$$|\theta^\kappa(t)|_{L^2} \leq |\theta^\kappa(T_j)|_{L^2}, \quad |\partial_x \theta^\kappa(t)|_{L^2} \leq |\partial_x \theta^\kappa(T_j)|_{L^2}, \quad |\partial_{xx} \theta^\kappa(t)|_{L^2} \leq |\partial_{xx} \theta^\kappa(T_j)|_{L^2}$$

for $t \in [T_j, T_{j+1}]$. Substituting these into the equations for $\partial_y \theta^\kappa$ and $\partial_{xy} \theta^\kappa$ and integrating gives

$$|\partial_y \theta^\kappa(t)|_{L^2} \leq N_j(t - T_j) |\partial_x \theta^\kappa(T_j)|_{L^2} + |\partial_y \theta^\kappa(T_j)|_{L^2}, \quad (4.5.3)$$

$$|\partial_{xy} \theta^\kappa(t)|_{L^2} \leq N_j(t - T_j) |\partial_{xx} \theta^\kappa(T_j)|_{L^2} + |\partial_{xy} \theta^\kappa(T_j)|_{L^2}, \quad (4.5.4)$$

for all $t \in [T_j, T_{j+1}]$. Using all preceding estimates in the ∂_{yy} -equation yields

$$|\partial_{yy} \theta^\kappa(t)|_{L^2} \leq (t - T_j)^2 N_j^2 |\partial_{xy} \theta^\kappa(T_j)|_{L^2} + (t - T_j) N_j^2 \frac{1}{\varepsilon_j} |\partial_x \theta^\kappa(T_j)|_{L^2} + |\partial_{yy} \theta^\kappa(T_j)|_{L^2}.$$

Taking the supremum over $[T_j, T_{j+1}]$ and using $N_j t_j \geq 1$,

$$\sup_{t \in [T_j, T_{j+1}]} |\theta^\kappa|_{\dot{H}^2} \leq (t_j^2 N_j^2 + 1) |\theta^\kappa(T_j)|_{H^2} + \frac{t_j}{\varepsilon_j} N_j^2 |\theta^\kappa(T_j)|_{H^1}.$$

This bound holds for every j . Setting $A_j = \sup_{t \in [T_j, T_{j+1}]} |\theta^\kappa|_{H^2}$ and substituting the explicit parameters, together with the bound $|\theta^\kappa|_{H^1}^2 \leq |\theta^\kappa|_{H^2}^2$, gives

$$A_j \leq (2^{2\alpha j} + 2) A_{j-1} + C_\alpha 2^{10j}.$$

Normalise so that $A_0 = 1$. The auxiliary sequence $B_j = A_j + C_\alpha 2^{10j}$ satisfies

$$B_j \leq 2C_\alpha 2^{10j} + (2^{2\alpha j} + 1)(B_{j-1} - C_\alpha 2^{10(j-1)}),$$

from which, for $j \geq J(\alpha)$,

$$B_j \leq (2^{2\alpha j} + 1)B_{j-1}, \quad \text{and hence} \quad B_j \leq \tilde{C}_\alpha 2^{\alpha j(j+1)}.$$

Returning to A_j ,

$$\sup_{t \in [T_j, T_{j+1}]} |\theta^\kappa|_{H^2} \leq \tilde{C}_\alpha \cdot 2^{\alpha j(j+1)} \quad \text{for all } j.$$

Comparing with the H^1 lower bound of Lemma 4.5.2 gives the claimed estimate $|\theta^\kappa|_{H^2} \leq C_\alpha |\theta|_{H^1}^2$, where θ is the inviscid solution. $\square$

4.5.3 Proof of Theorem 3

The previous lemmas establish anomalous dissipation for initial data given by a pure harmonic. To conclude the proof of Theorem 3, it remains to treat the case of a small L^2 perturbation and to verify that the velocity field $u(t)$ defined in (4.5.1) belongs to $L^1([0, 1]; C^{\alpha'})$ for any $\alpha' < 1$. For the latter, we simply compute that

$$|u|_{L^1([0, 1]; C^{\alpha'})} \lesssim \sum_{j \geq J(\alpha)} t_j N_j^{\alpha'} \lesssim \sum_{j \geq J(\alpha)} 2^{(1+\alpha)\alpha' j - j} < +\infty$$

once we take $0 < \alpha < \frac{1}{\alpha'} - 1$.

Suppose θ_0 satisfies the proximity condition of Theorem 3, with $\varepsilon_\alpha > 0$ to be chosen. Without loss of generality, take $\psi = \sin(Mx)$ and $\lambda = 1$, with ψ orthogonal to $\theta_0 - \psi$ in L^2 . Decompose the initial datum as

$$\theta_0(x, y) = \sin(Mx) + (\theta_0 - \sin(Mx)) \stackrel{\text{def}}{=} \theta_0^L + \theta_0^H,$$

so $|\theta_0|_{L^2}^2 = |\theta_0^L|_{L^2}^2 + |\theta_0^H|_{L^2}^2$. The smallness assumption gives

$$\sqrt{1 - \varepsilon_\alpha^2} |\theta_0^H|_{L^2} \leq \varepsilon_\alpha |\theta_0^L|_{L^2},$$

hence $|\theta_0|_{L^2} \geq (1 - \frac{\varepsilon_\alpha}{\sqrt{1 - \varepsilon_\alpha^2}}) |\theta_0^L|_{L^2}$. The preceding lemmas applied to the “low-frequency” part $\theta_0^L = \sin(Mx)$ give, for any $0 < \kappa \leq 1$,

$$\frac{1}{2} (|\theta_0^L|_{L^2}^2 - |\theta^{L, \kappa}(1)|_{L^2}^2) = \kappa \int_0^1 |\nabla \theta^{L, \kappa}|_{L^2}^2 dt \geq \chi_\alpha |\theta_0^L|_{L^2}^2,$$

where $\theta^{L, \kappa}$ solves the advection-diffusion equation with initial data θ_0^L . Letting $\theta^{H, \kappa}$ denote the solution with initial data θ_0^H , linearity gives $\theta^\kappa = \theta^{L, \kappa} + \theta^{H, \kappa}$. Combining the energy estimates for both parts,

$$\begin{aligned} |\theta^\kappa(1)|_{L^2} &\leq |\theta^{L, \kappa}(1)|_{L^2} + |\theta^{H, \kappa}(1)|_{L^2} \leq (1 - 2\chi_\alpha) |\theta_0^L|_{L^2} + |\theta_0^H|_{L^2} \\ &\leq \left((1 - 2\chi_\alpha) \left(1 - \frac{\varepsilon_\alpha}{\sqrt{1 - \varepsilon_\alpha^2}}\right)^{-1} + \frac{\varepsilon_\alpha}{\sqrt{1 - \varepsilon_\alpha^2}} \right) |\theta_0|_{L^2} < \left(1 - \frac{1}{10} \chi_\alpha\right) |\theta_0|_{L^2}, \end{aligned}$$

provided $\varepsilon_\alpha = \chi_\alpha/100$. The energy identity (4.0.2) then yields the required lower bound on $\kappa \int_0^1 |\nabla \theta^\kappa|_{L^2}^2$. $\square$

4.5.4 Proof of Theorem 4

Theorem 4 establishes anomalous dissipation for arbitrary mean-zero initial data $\theta_0 \in H^2(\mathbb{T}^2)$. As in the proof of Theorem 3, the argument proceeds via Proposition 4.2.1, but with a velocity field that now depends on θ_0 . This time, given $\alpha > 0$, we take

$$t_j = 2^{-j}, \quad N_j = 2^{(1+\alpha)j}, \quad \varepsilon_j = a_0 \exp\left(-30\left(1 + \frac{1}{2^\alpha - 1}\right)\right) \cdot 2^{-2j},$$

and define the velocity field $u(t)$ for $t \in [T_j, T_{j+1})$ by

$$u(t, x, y) = \begin{cases} \begin{pmatrix} (-1)^{s_j} S_{\varepsilon_j}(N_j y) \\ 0 \end{pmatrix} & j \text{ even,} \\ \begin{pmatrix} 0 \\ (-1)^{s_j} S_{\varepsilon_j}(N_j x) \end{pmatrix} & j \text{ odd,} \end{cases} \quad (4.5.5)$$

where $T_j = \sum_{i=0}^j t_i$ with $t_0 = 0$. The amplitude $a_0 \in (0, 1]$ and the signs $s_j \in \{0, 1\}$ are chosen adaptively based on the initial data in a manner described below. Structurally, the velocity field is the same alternating-shear construction as in (4.5.1).

The verification of Proposition 4.2.1 follows the same strategy as in the proof of Theorem 3. The only modification is in establishing the H^1 lower bound, since the initial data is now general. The sign s_j is selected at each step to ensure sufficient growth. Specifically, for j odd, introduce the two candidate iterates $\theta_{j+1}^\pm = \theta_j(x \pm t_j S_{\varepsilon_j}(N_j y), y)$ and differentiate:

$$\partial_{i_{j+1}} \theta_{j+1}^\pm(x, y) = \partial_{i_{j+1}} \theta_j(x \pm t_j S_{\varepsilon_j}(N_j y), y) \pm t_j N_j S'_{\varepsilon_j}(N_j y) \partial_{i_j} \theta_j(x \pm t_j S_{\varepsilon_j}(N_j y), y).$$

Since the cross terms vanish when summing over the two signs,

$$\sum_{\pm} |\partial_{i_{j+1}} \theta_{j+1}^\pm|_{L^2}^2 = 2|\partial_{i_{j+1}} \theta_j|_{L^2}^2 + 2(t_j N_j)^2 |S'_{\varepsilon_j}(N_j y) \partial_{i_j} \theta_j|_{L^2}^2.$$

At least one of the two choices of sign must therefore satisfy

$$|\partial_{i_{j+1}} \theta_{j+1}^{(-1)^{s_j}}|_{L^2} \geq t_j N_j |S'_{\varepsilon_j}(N_j y) \partial_{i_j} \theta_j|_{L^2} \geq t_j N_j |\partial_{i_j} \theta_j|_{L^2} \left(1 - \sqrt{\varepsilon_j} \frac{|\theta_j|_{W^{1,\infty}}}{|\partial_{i_j} \theta_j|_{L^2}}\right);$$

this selects s_j . In the notation of the bootstrap scheme (with $\alpha = 1$ for simplicity), the ratio A_j now satisfies

$$A_j \leq 2^{-j} (1 - \sqrt{\varepsilon_j} R_j)^{-1},$$

while the inequality for R_{j+1} is unchanged. Choosing $a_0 > 0$ sufficiently small depending only on θ_0 ensures $A_1 < 1$, at which point the bootstrap closes exactly as before and yields the desired H^1 lower bound. $\square$

4.6 Non-Uniqueness of Weak Solutions

The goal of this section is to prove Theorem 5, following [DEIJ22]. Recall that $\theta \in C_w(0, T; L^2(\mathbb{T}^d))$ is called a weak solution of the transport equation (4.3.4) on $\mathbb{T}^d \times [0, T]$ if

$$\int_0^T \int_{\mathbb{T}^d} \theta (\partial_t \varphi + u \cdot \nabla \varphi) dx dt = - \int_{\mathbb{T}^d} \theta_0(x) \varphi(x, 0) dx \quad (4.6.1)$$

holds, for all test functions $\varphi \in C_0^\infty([0, T] \times \mathbb{T}^d)$.

Proof of Theorem 5. Two distinct weak solutions are constructed: one is time-irreversible and arises as a vanishing viscosity limit, while the other is obtained by time-reversal.

Time irreversible weak solution: Define θ as the weak limit, as $\kappa_k \rightarrow 0$, of the solutions θ^{κ_k} of the advection-diffusion equation with velocity u_* . The uniform L^∞ bound on θ^{κ_k} guarantees — via the Aubin-Lions lemma — convergence in $C([0, 2T]; w-L^2(\mathbb{T}^d))$, where $w-L^2$ denotes L^2 with the weak topology. Since the transport equation is linear and $u \in L^\infty(0, 2T; L^\infty(\mathbb{T}^d))$, passing to the limit shows that θ satisfies the weak formulation (4.6.1) on $[0, 2T] \times \mathbb{T}^d$. Weak lower semicontinuity of the L^2 norm then gives, for all $t \in [T, 2T]$,

$$|\theta(t)|_{L^2}^2 \leq \liminf_{\kappa \rightarrow 0} |\theta^\kappa(t)|_{L^2}^2 \leq |\theta_0|_{L^2}^2 - \limsup_{\kappa \rightarrow 0} \kappa \int_0^T |\nabla \theta^\kappa|_{L^2}^2 dt \leq (1 - \chi_\alpha) |\theta_0|_{L^2}^2 < |\theta_0|_{L^2}^2 \quad (4.6.2)$$

upon applying Theorem 4. Thus,

$$\sup_{t \in [T, 2T]} |\theta(t)|_{L^2}^2 \leq (1 - \chi_\alpha) |\theta_0|_{L^2}^2 < |\theta_0|_{L^2}^2, \quad (4.6.3)$$

and hence the inviscid limit has strictly lost L^2 mass over $[0, T]$.

Time reversible weak solution: A second weak solution $\bar{\theta}$ is produced by time-reversal: define

$$\bar{\theta}(t) = \begin{cases} \theta(t) & t \in [0, T], \\ \theta(2T - t) & t \in [T, 2T]. \end{cases} \quad (4.6.4)$$

Since $\theta \in C_w([0, T]; \mathbb{T}^d)$, this formula gives $\bar{\theta} \in C_w([0, 2T]; \mathbb{T}^d)$. To verify that $\bar{\theta}$ satisfies (4.6.1) on $[0, 2T]$, take any $\varphi \in C_0^\infty([0, 2T] \times \mathbb{T}^d)$ and check:

$$\int_0^{2T} \int_{\mathbb{T}^d} \bar{\theta}(x, t) \partial_t \varphi(x, t) dx dt + \int_0^{2T} \int_{\mathbb{T}^d} \bar{\theta}(x, t) u(x, t) \cdot \nabla \varphi(x, t) dx dt = - \int_{\mathbb{T}^d} \theta_0(x) \varphi(x, 0) dx. \quad (4.6.5)$$

Decompose the test function into its even and odd parts about $t = T$:

$$\varphi = \varphi_e + \varphi_o, \quad \varphi_{e/o} \stackrel{\text{def}}{=} \frac{\varphi(T + (t - T)) \pm \varphi(T - (t - T))}{2}. \quad (4.6.6)$$

Because $\bar{\theta}(x, t)$ is even and $u(x, t)$ is odd about $t = T$, the contribution from φ_e vanishes, and the equation reduces to

$$\int_0^{2T} \int_{\mathbb{T}^d} \bar{\theta}(x, t) \partial_t \varphi_o(x, t) \, dx \, dt + \int_0^{2T} \int_{\mathbb{T}^d} \bar{\theta}(x, t) u(x, t) \cdot \nabla \varphi_o(x, t) \, dx \, dt = - \int_{\mathbb{T}^d} \theta_0(x) \varphi(x, 0) \, dx. \quad (4.6.7)$$

Splitting the time integral at $t = T$ gives

$$\begin{aligned} \int_0^T \int_{\mathbb{T}^d} \theta(x, t) \partial_t \varphi_o(x, t) \, dx \, dt + \int_0^T \int_{\mathbb{T}^d} \theta(x, t) u_*(x, t) \cdot \nabla \varphi_o(x, t) \, dx \, dt &= - \int_{\mathbb{T}^d} \theta_0(x) \varphi(x, 0) \, dx \\ - \int_T^{2T} \int_{\mathbb{T}^d} \theta(x, 2T - t) \partial_t \varphi_o(x, t) \, dx \, dt + \int_T^{2T} \int_{\mathbb{T}^d} \theta(x, 2T - t) u_*(x, 2T - t) \cdot \nabla \varphi_o(x, t) \, dx \, dt. \end{aligned}$$

The change of variables $\tau = 2T - t$ converts the integral over $[T, 2T]$ into one over $[0, T]$. Setting $\psi(x, \tau) = \varphi_o(x, 2T - \tau)$ — which satisfies $\psi(x, T) = 0$ since φ_o is odd about T — yields

$$\begin{aligned} \int_0^T \int_{\mathbb{T}^d} \theta(x, t) \partial_t \varphi_o(x, t) \, dx \, dt + \int_0^T \int_{\mathbb{T}^d} \theta(x, t) u_*(x, t) \cdot \nabla \varphi_o(x, t) \, dx \, dt &= - \int_{\mathbb{T}^d} \theta_0(x) \varphi(x, 0) \, dx \\ + \int_0^T \int_{\mathbb{T}^d} \theta(x, \tau) \partial_\tau \psi(x, \tau) \, dx \, d\tau + \int_0^T \int_{\mathbb{T}^d} \theta(x, \tau) u_*(x, \tau) \cdot \nabla \psi(x, \tau) \, dx \, d\tau. \end{aligned}$$

Applying the weak formulation for θ on $[0, T]$ with test function ψ (using $\psi(x, T) = 0$ and $\psi(x, 0) = \varphi_o(x, 2T)$) gives

$$\int_0^T \int_{\mathbb{T}^d} [\theta(x, \tau) \partial_\tau \psi(x, \tau) + \theta(x, \tau) u_*(x, \tau) \cdot \nabla \psi(x, \tau)] \, dx \, d\tau = - \int_{\mathbb{T}^d} \theta_0(x) \varphi_o(x, 2T) \, dx.$$

The weak formulation of θ on $[0, T]$ with test function φ_o also gives

$$\int_0^T \int_{\mathbb{T}^d} [\theta(x, t) \partial_t \varphi_o(x, t) + \theta(x, t) u_*(x, t) \cdot \nabla \varphi_o(x, t)] \, dx \, dt = - \int_{\mathbb{T}^d} \theta_0(x) \varphi_o(x, 0) \, dx.$$

Adding the two equations and using

$$\varphi_o(x, 0) = -\varphi_o(x, 2T) = \frac{1}{2} \varphi(x, 0) \quad (4.6.8)$$

confirms that $\bar{\theta}$ satisfies (4.6.1) on $[0, 2T]$, so $\bar{\theta} \in C_w([0, 2T]; \mathbb{T}^d)$ is a weak solution. To see that $\bar{\theta} \neq \theta$, note that

$$|\bar{\theta}(2T)|_{L^2}^2 = |\theta_0|_{L^2}^2. \quad (4.6.9)$$

In light of (4.6.3), the solutions θ and $\bar{\theta}$ are distinct. $\square$

4.7 Discussion: Obukhov-Corrsin Theory

In the context of passive scalar turbulence, Obukhov [Obu49] and Corrsin [Cor51] studied the ‘inertial-range’ scaling behavior of scalar structure functions $S_p^\theta(\ell) \stackrel{\text{def}}{=} \langle |\delta_\ell \theta|^p \rangle \sim \ell^{\zeta_p(\theta)}$ in a fully developed homogenous isotropic velocity field exhibiting Kolmogorov 1941 (K41) ‘monofractal’ scaling [Kol41]

$$S_p^u(\ell) \stackrel{\text{def}}{=} \langle |\delta_\ell u|^p \rangle \sim (\varepsilon \ell)^{p/3}, \quad \ell_\nu \ll \ell \ll L^u \quad (4.7.1)$$

for all $p \geq 1$ where L^u is the integral scale of the velocity field and ℓ_ν is the dissipation scale (the K41 prediction being $\ell_\nu = (\nu^3/\varepsilon)^{1/4}$ where $\varepsilon \stackrel{\text{def}}{=} \lim_{\nu \rightarrow 0} \nu \langle |\nabla u^\nu|^2 \rangle > 0$ is the anomalous energy dissipation rate). Said another way, in the idealized limit $\nu \rightarrow 0$, the velocity field is assumed to be 1/3-Hölder and not better. Based on dimensional grounds, Obukhov and Corrsin independently predicted that the scalar field would also exhibit the same scaling

$$S_p^\theta(\ell) \stackrel{\text{def}}{=} \langle |\delta_\ell \theta|^p \rangle \sim (\chi/\varepsilon^{1/3})^{p/2} \ell^{p/3}, \quad \ell_\nu \lesssim \ell_\kappa \ll \ell \ll L^\theta \lesssim L^u \quad (4.7.2)$$

where $\chi \stackrel{\text{def}}{=} \lim_{\kappa, \nu \rightarrow 0} \kappa \langle |\nabla \theta^\kappa|^2 \rangle > 0$ is the (presumed) anomalous dissipation of the passive scalar, L^θ is the typical length-scale of the scalar input initially or by a force, and ℓ_κ is the dissipative length for the scalar field ($\ell_\kappa = (\kappa^3/\varepsilon)^{1/4}$ in the Corrsin-Obukhov theory). Their scaling theory can be generalized as

$$S_p^u(\ell) \sim \ell^{\alpha p}, \quad \alpha \in (0, 1) \quad \text{implies} \quad S_p^\theta(\ell) \sim \ell^{(\frac{1-\alpha}{2})p}. \quad (4.7.3)$$

In the idealized limit of $\nu, \kappa \rightarrow 0$, this says that if the velocity $u \in C^\alpha$ is Hölder with exponent $\alpha \in (0, 1)$ and not better, then the scalar should be Hölder $\theta \in C^\beta$ with exponent $\beta = (1-\alpha)/2$ and not better. These constraints can be understood as a consequence of the fractal geometry of scalar level sets in rough velocities [CP93, CP94]. Moreover, the entire picture has been generalized to accommodate (the more realistic setting) of multifractal velocity fields with the property that $S_p^u(\ell) \sim \ell^{\zeta_p(u)}$ where $\zeta_p(u)$ may depend non-linearly on p resulting in constraints on the multifractal spectrum of the scalar $\zeta_p(\theta)$ [Eyi96].

In analogy to the Onsager conjecture for the dissipation anomaly of kinetic energy in incompressible fluids [Ons49], one can regard the above theory as setting a threshold condition for the anomalous dissipation of scalar energy [Eyi96]. Namely, if $u \in C^\alpha$ and $\theta^\kappa \in C^\beta$ uniformly then

$$\chi \stackrel{\text{def}}{=} \kappa \int_0^T \int_{\mathbb{T}^d} |\nabla \theta^\kappa|^2 dx dt \rightarrow 0 \quad \text{unless} \quad \beta > \frac{1-\alpha}{2}. \quad (4.7.4)$$

Along these lines, we first establish an upper bound on the dissipation for vanishing diffusion limits in rough velocity fields. A similar estimate was provided for viscous energy dissipation in the context of Onsager’s conjecture for hydrodynamic turbulence [DE19].

We also study what happens when the velocity field is smooth up until a single point in time where it may lose regularity. The latter is relevant to the problem in which an inertial range for the velocity field evolves dynamically by some cascade process to the point where the field becomes non-smooth in a way consistent with the observed long-time inertial range scaling in real turbulence. In fact, one has the following result.

Theorem 6 ([DEIJ22, Theorem 1.4]). *Let $u \in L^1([0, T]; C^\alpha(\mathbb{T}^d))$ for $\alpha \in (0, 1]$ be a given divergence free vector field. Suppose that the family $\{\theta^\kappa\}_{\kappa>0}$ is bounded in $L^\infty([0, T]; C^\beta(\mathbb{T}^d))$ for $\beta \in (0, 1]$ uniformly in κ , then*

$$\kappa \int_0^T \int_{\mathbb{T}^d} |\nabla \theta^\kappa|^2 dx dt \leq C \kappa^{\frac{\alpha+2\beta-1}{\alpha+1}} \quad (4.7.5)$$

for an absolute constant C depending only on T and the Hölder norms of the solutions. In particular, if $\beta > (1 - \alpha)/2$ then there can be no anomalous scalar dissipation. If furthermore

$$u \in L^1_{loc}([0, T]; W^{1,\infty}(\mathbb{T}^d)) \cap L^1([0, T]; C^\alpha(\mathbb{T}^d)) \quad \text{for } \alpha < 1,$$

and if $\beta = (1 - \alpha)/2$, then

$$\lim_{\kappa \rightarrow 0} \kappa \int_0^T \int_{\mathbb{T}^d} |\nabla \theta^\kappa|^2 dx dt = 0. \quad (4.7.6)$$

Proof. Let $\bar{f}_\ell = \varphi_\ell * f$ for any $\ell > 0$. Mollifying the equations, one finds

$$\partial_t (\overline{\theta^\kappa})_\ell + \bar{u}_\ell \cdot \nabla (\overline{\theta^\kappa})_\ell = \kappa \Delta (\overline{\theta^\kappa})_\ell - \nabla \cdot \tau_\ell(u, \theta^\kappa) \quad (4.7.7)$$

where $\tau_\ell(f, g) = \overline{(fg)}_\ell - \bar{f}_\ell \bar{g}_\ell$. A straightforward calculation for any $0 \leq t \leq T$ shows that

$$\begin{aligned} \kappa \int_t^T \int_{\mathbb{T}^d} |\nabla \theta^\kappa|^2 dx dt' &= \int_t^T \int_{\mathbb{T}^d} \nabla (\overline{\theta^\kappa})_\ell \cdot \tau_\ell(u, \theta^\kappa) dx dt' + \kappa \int_t^T \int_{\mathbb{T}^d} |\nabla (\overline{\theta^\kappa})_\ell|^2 dx dt' \\ &\quad + \frac{1}{2} \int_{\mathbb{T}^d} \tau_\ell(\theta^\kappa(t), \theta^\kappa(t)) dx - \frac{1}{2} \int_{\mathbb{T}^d} \tau_\ell(\theta^\kappa(T), \theta^\kappa(T)) dx. \end{aligned} \quad (4.7.8)$$

Using standard estimates for mollified gradients and the Constantin-E-Titi [CET94] commutator estimate

$$|\nabla \bar{f}_\ell|_{L^\infty} \leq |\theta|_{C^\alpha} \ell^{\alpha-1}, \quad |\tau_\ell(f, g)|_{L^\infty} \leq |\theta|_{C^\alpha} |g|_{C^\beta} \ell^{\alpha+\beta}, \quad f \in C^\alpha, g \in C^\beta \quad (4.7.9)$$

together with the fact that $\tau_\ell(f, f) \geq 0$ we arrive at an upper bound for the scalar dissipation

$$\kappa \int_t^T \int_{\mathbb{T}^d} |\nabla \theta^\kappa|^2 dx dt' \leq \ell^{\alpha+2\beta-1} |\theta^\kappa|_{L_t^\infty C_x^\beta}^2 |u|_{L^1(t, T; C_x^\alpha)} + (\kappa(T-t)\ell^{2(\beta-1)} + \ell^{2\beta}) |\theta^\kappa|_{L_t^\infty C_x^\beta}^2. \quad (4.7.10)$$

Setting $t = 0$ and optimizing ℓ as a function of κ we find $\ell = \kappa^{1/(\alpha+1)}$ and (4.7.5) follows. The second statement of the theorem follows by dividing the time interval into $[0, T - \varepsilon] \times [T - \varepsilon, T]$ and using the assumed C^1 regularity of u on the interval $[0, T - \varepsilon]$ together with the uniform Hölder on the entire interval $[0, T]$ and the fact that ε is allowed arbitrarily small (and can vanish as $\kappa \rightarrow 0$). $\square$

In light of Theorems 3 and 6, Drivas et al. [DEIJ22] raise the following open question.

Question 4.7.1. *Fix $\alpha \in (0, 1)$. Does there exist divergence-free vector field*

$$u \in L^1([0, 1]; C^\alpha(\mathbb{T}^d))$$

such that $\{\theta^\kappa\}_{\kappa>0}$ is bounded in $L^\infty([0, T]; C^\beta(\mathbb{T}^d))$ for every $\beta < (1 - \alpha)/2$, and

$$\liminf_{\kappa \rightarrow 0} \kappa \int_0^T \int_{\mathbb{T}^d} |\nabla \theta^\kappa|^2 dx dt > 0 ?$$

Drivas et al. [DEIJ22] also comment briefly on the nonlinear problem of establishing anomalous dissipation for solutions of the Navier-Stokes equations

$$\begin{aligned} \partial_t u^\nu + u^\nu \cdot \nabla u^\nu &= -\nabla p^\nu + \nu \Delta u^\nu, \\ \nabla \cdot u^\nu &= 0. \end{aligned}$$

As for passive scalars, experimental and numerical observations of hydrodynamic turbulence suggest that kinetic energy dissipation is non-vanishing in the limit of zero viscosity [Sre84, Sre98, PKW02, KIY⁺03], i.e. there exists $\varepsilon > 0$ independent of ν such that, in turbulent regimes, a family of Leray-Hopf solutions $\{u^\nu\}_{\nu>0}$ satisfies

$$\nu \int_0^T \int |\nabla u^\nu(x, t)|^2 dx dt \geq \varepsilon > 0. \quad (4.7.11)$$

This phenomenon of anomalous dissipation is so fundamental to our modern understanding of turbulence that it is often termed the “zeroth law”. In 1949 [Ons49], Lars Onsager offered significant insight into this phenomena in asserting that it requires that, at high Reynolds number, flow develop structures approximating singular ones with Hölder exponents not exceeding $1/3$. This assertion has since been proved [Eyi94, CET94] and dissipative weak solutions of the Euler equations with lower regularity have been constructed in a series of works using convex integration [DLS09, DLS10, DLS12, Ise18, BDSV19] and culminating in a construction of non-conservative solutions in the class $C_t C_x^{1/3-}$ by P. Isett. However, to this day, none of these constructions are achieved as zero viscosity limits of Navier-Stokes solutions obeying a physical energy balance (e.g. Leray-Hopf weak solutions). In [DEIJ22], Drivas, Elgindi, Iyer, and Jeong solved an analogous problem for passive scalars in a setting which models the effect of a finite-time singularity in an inviscid problem on anomalous dissipation in the corresponding viscous problem. Their result follows from a sufficient condition for anomalous dissipation assuming that the inviscid solution becomes singular in a controlled way. It is possible that one could deduce anomalous dissipation in the vanishing viscosity limit of Navier-Stokes solutions under some conditions on a (hypothetical) blowup in the Euler equation.

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